

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

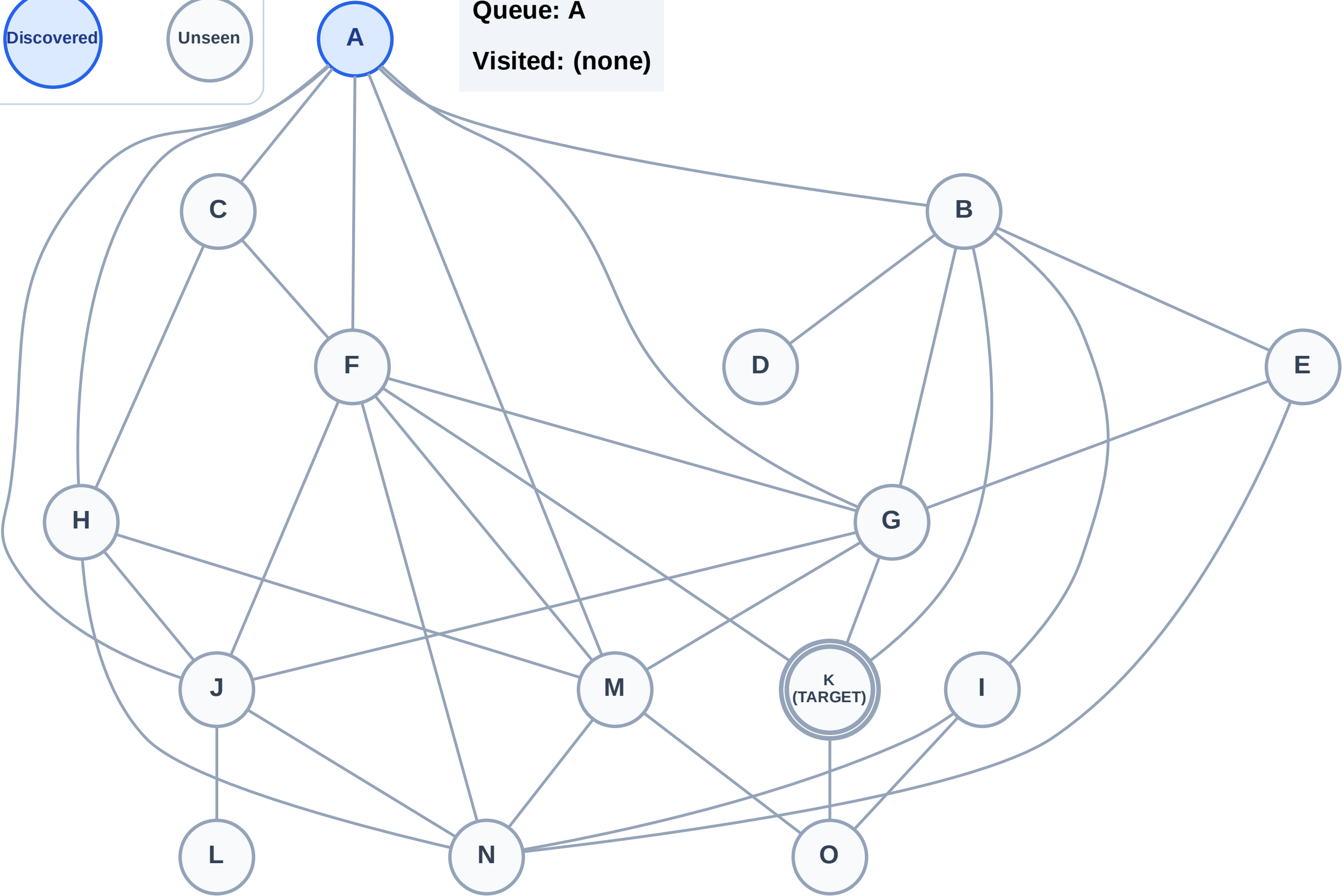
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

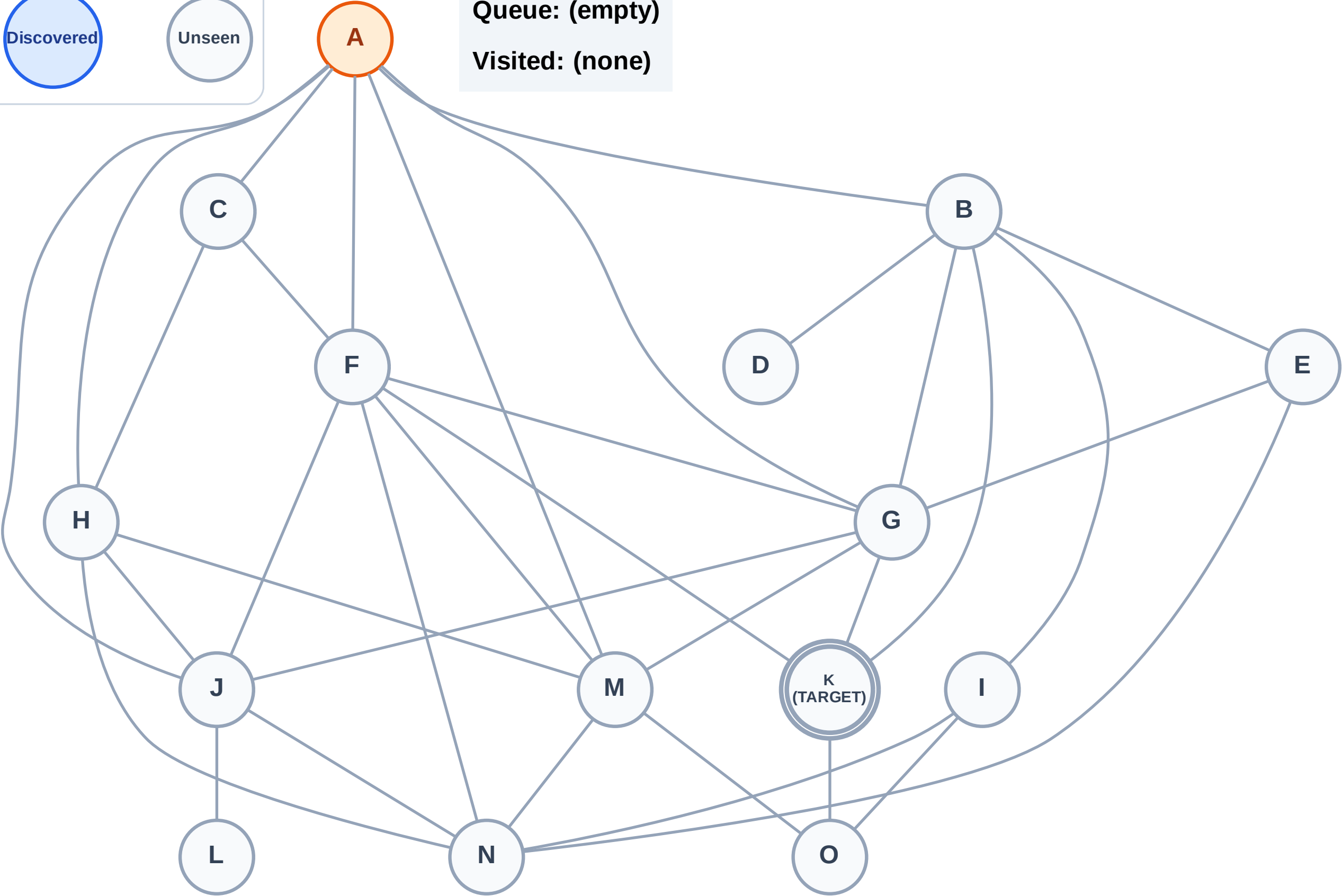
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, F, G, H, J, M from A

Legend

Complete

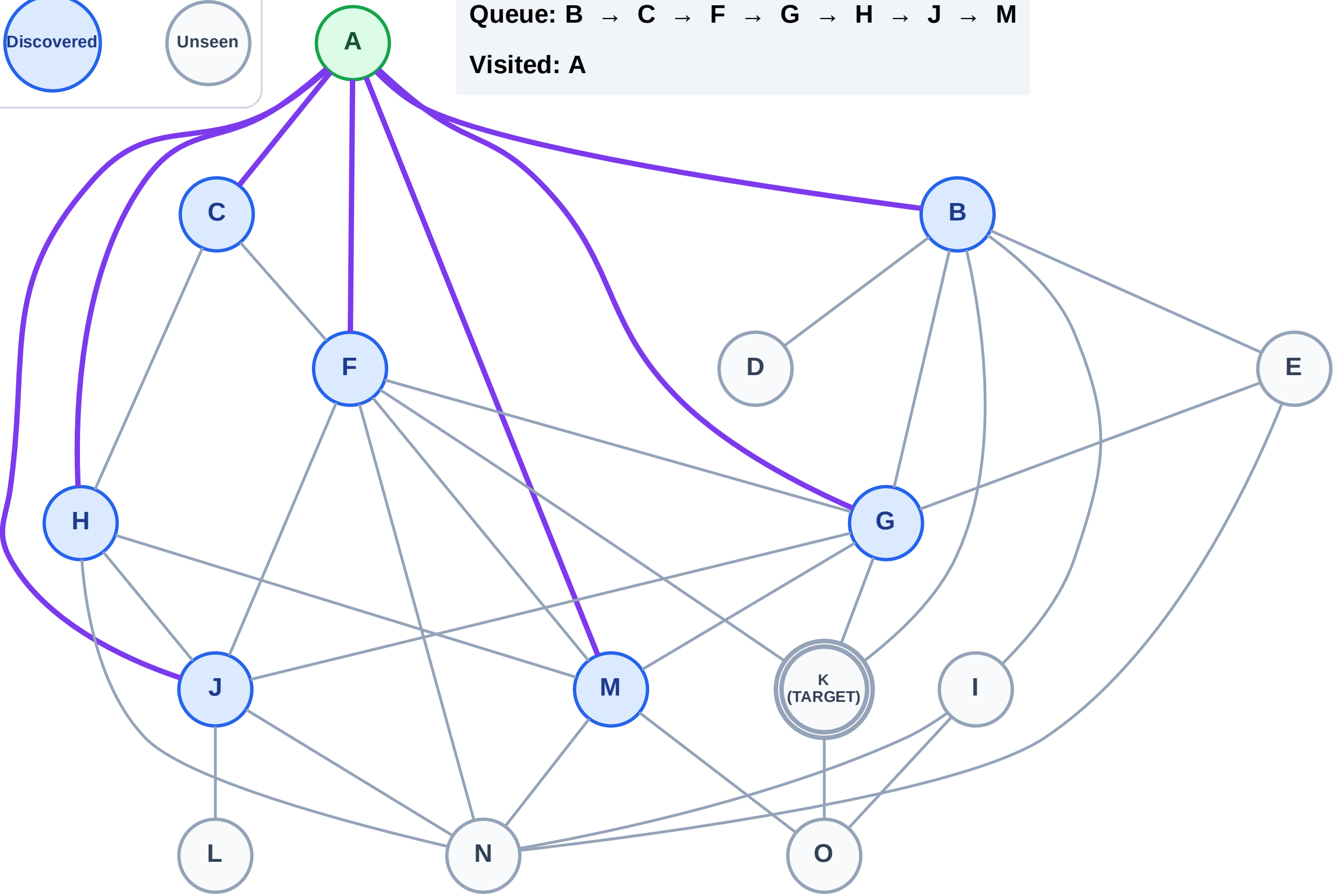
Processing

Discovered

Unseen

Queue: B → C → F → G → H → J → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

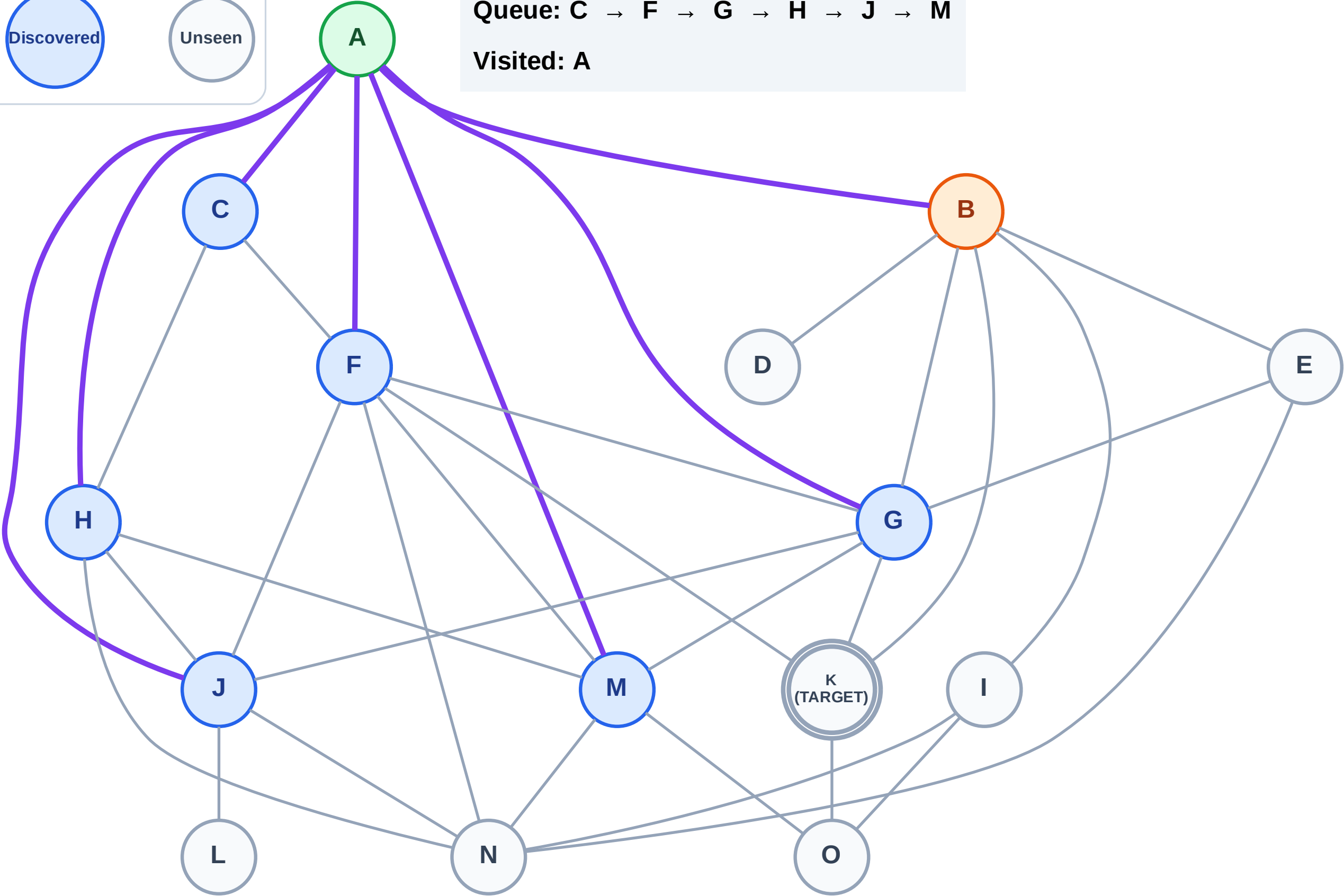
Processing

Discovered

Unseen

Queue: C → F → G → H → J → M

Visited: A



BFS Traversal

Step 5: Discover D, E, I, K from B

Legend

Complete

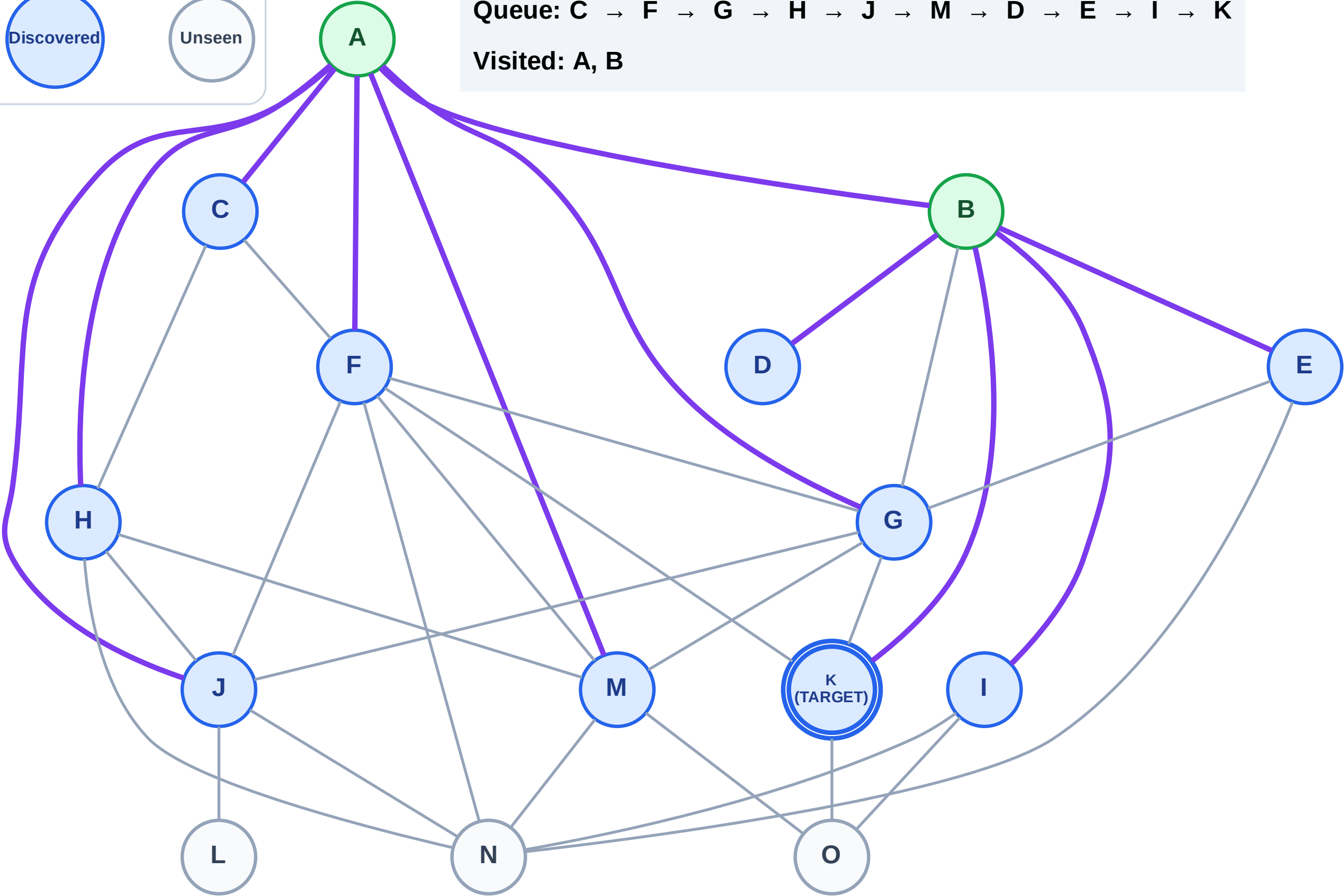
Processing

Discovered

Unseen

Queue: C → F → G → H → J → M → D → E → I → K

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

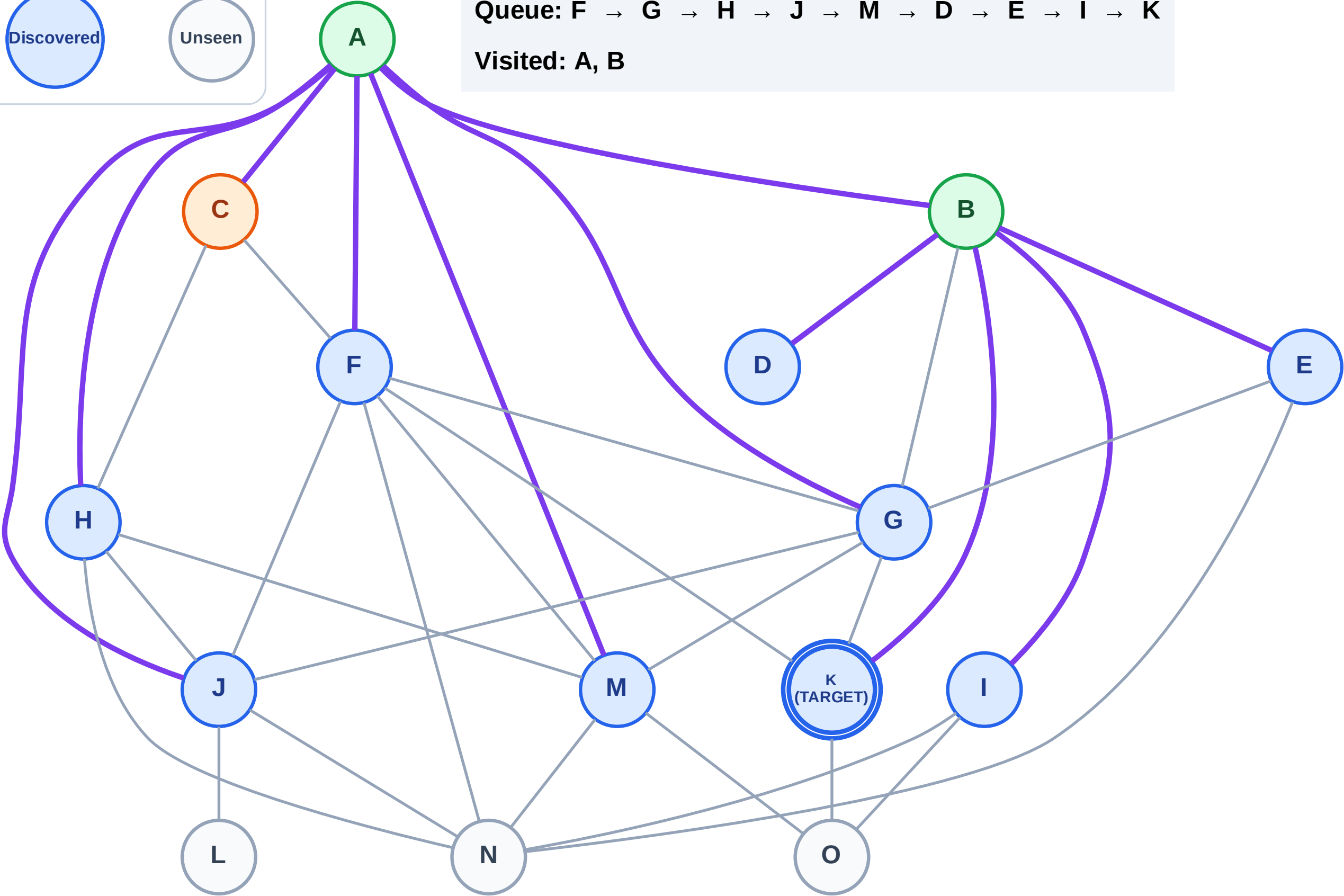
Processing

Discovered

Unseen

Queue: F → G → H → J → M → D → E → I → K

Visited: A, B



BFS Traversal

Step 7: No new neighbors from C

Legend

Complete

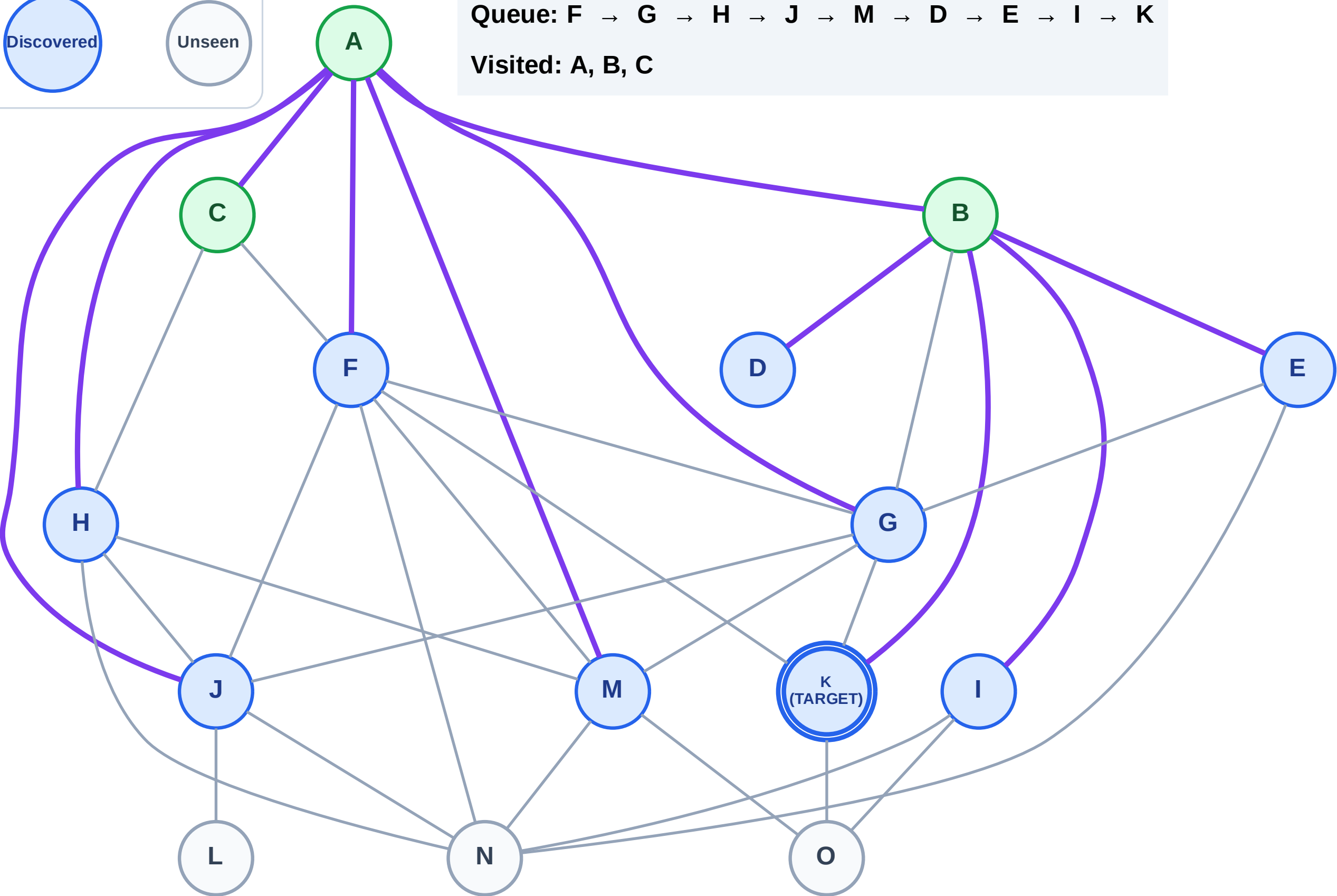
Processing

Discovered

Unseen

Queue: F → G → H → J → M → D → E → I → K

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

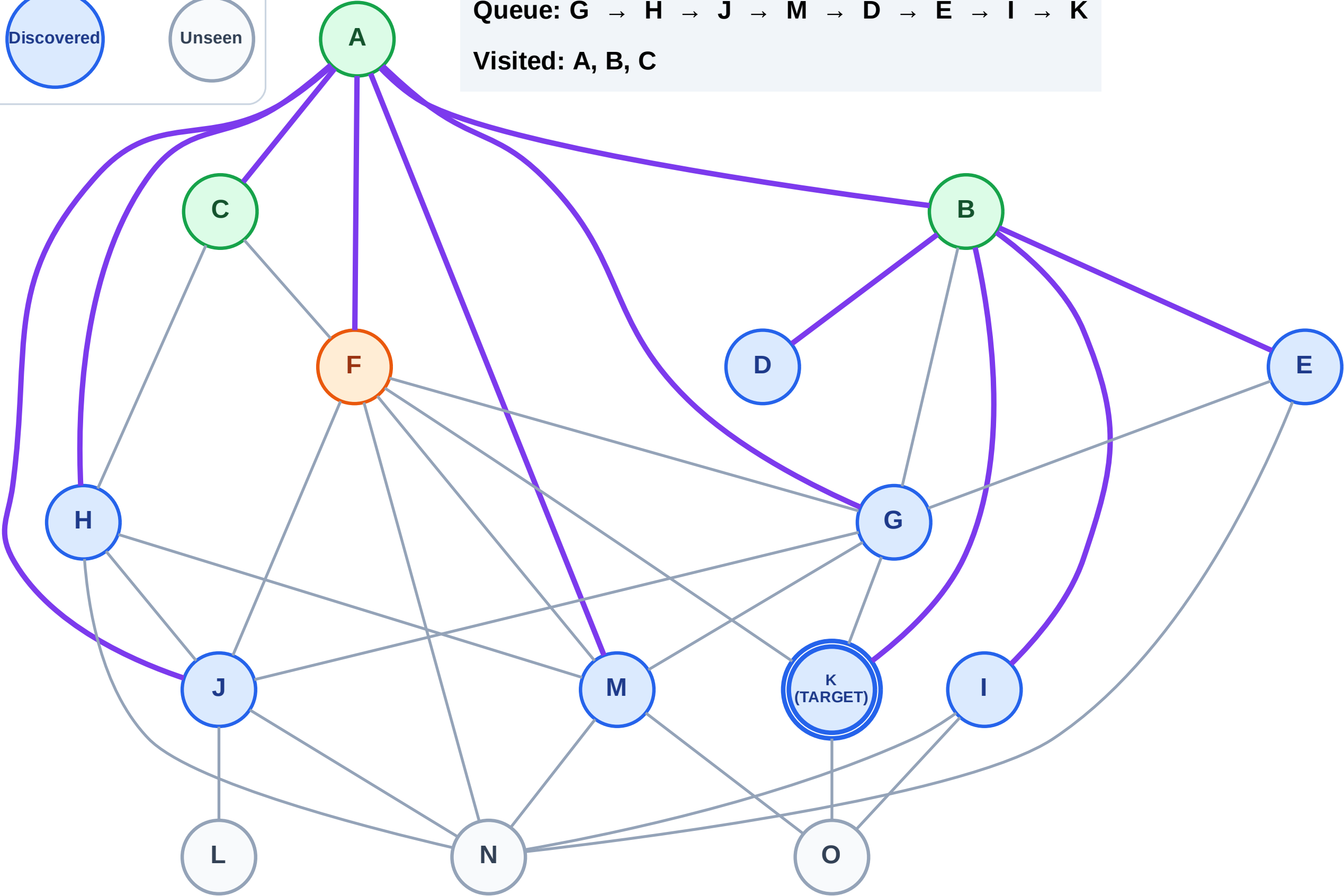
Processing

Discovered

Unseen

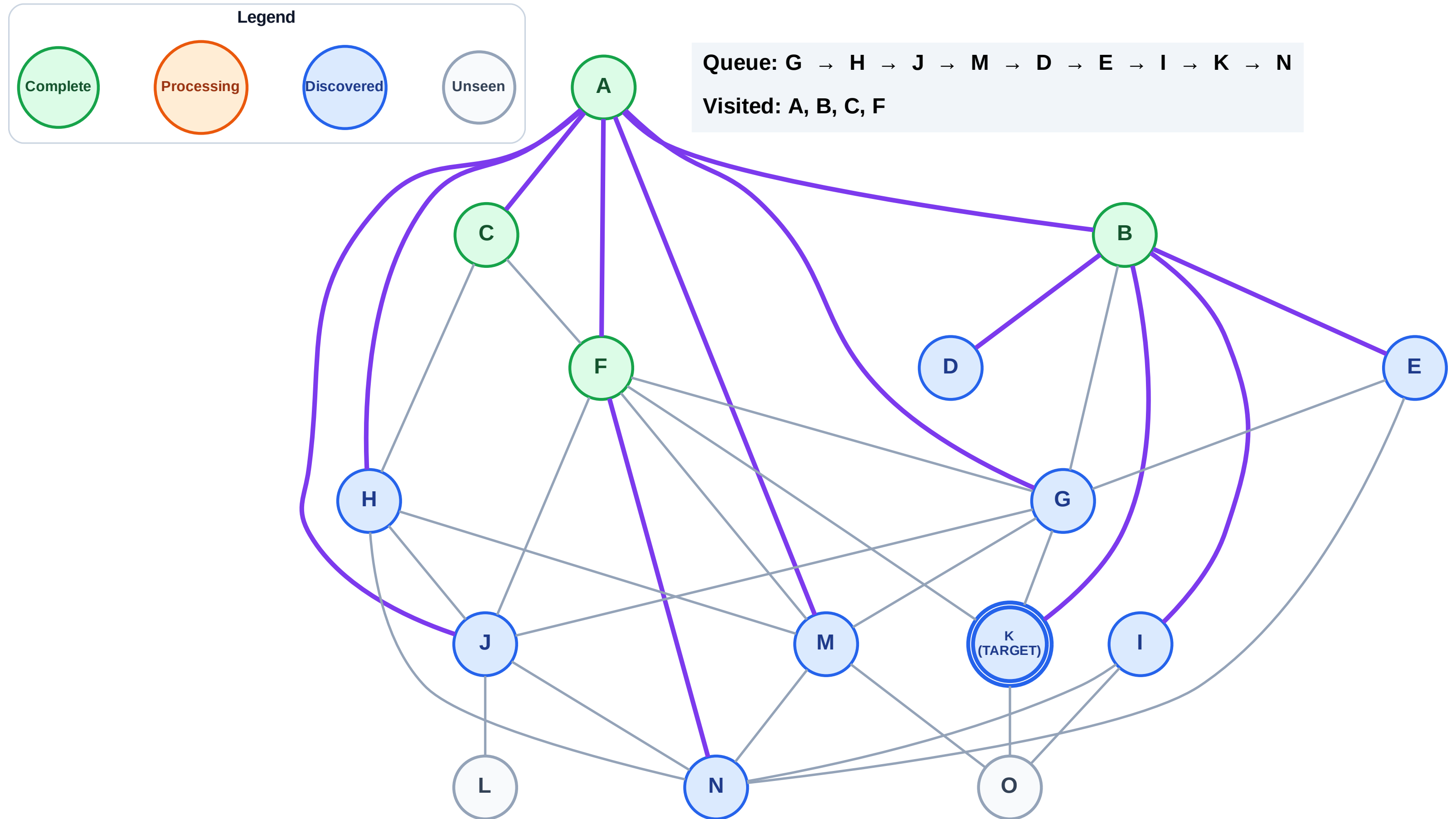
Queue: G → H → J → M → D → E → I → K

Visited: A, B, C



Step 9: Discover N from F

Step 9: Discover N from F



BFS Traversal

Step 10: Process node G

Legend

Complete

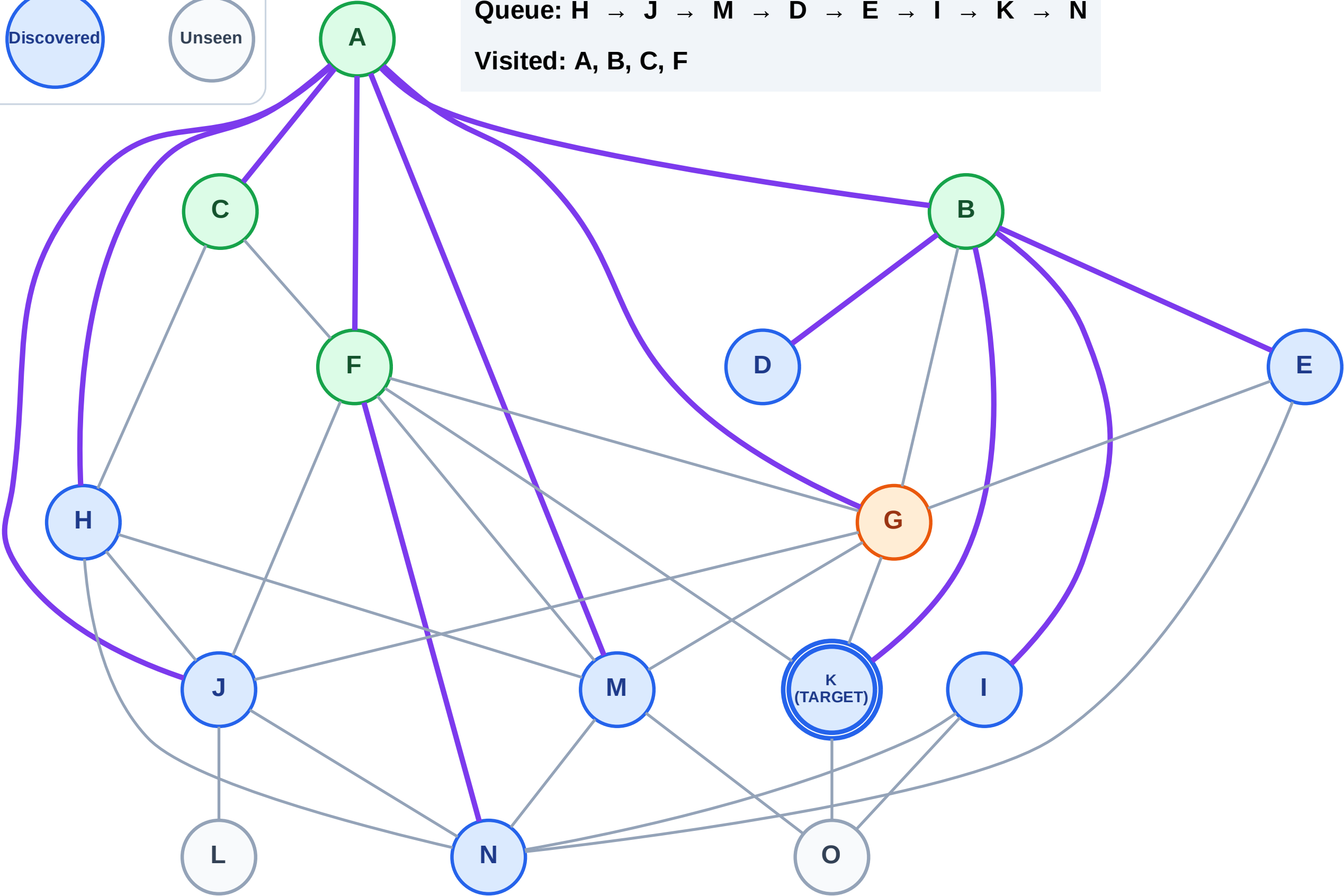
Processing

Discovered

Unseen

Queue: H → J → M → D → E → I → K → N

Visited: A, B, C, F



BFS Traversal

Step 11: No new neighbors from G

Legend

Complete

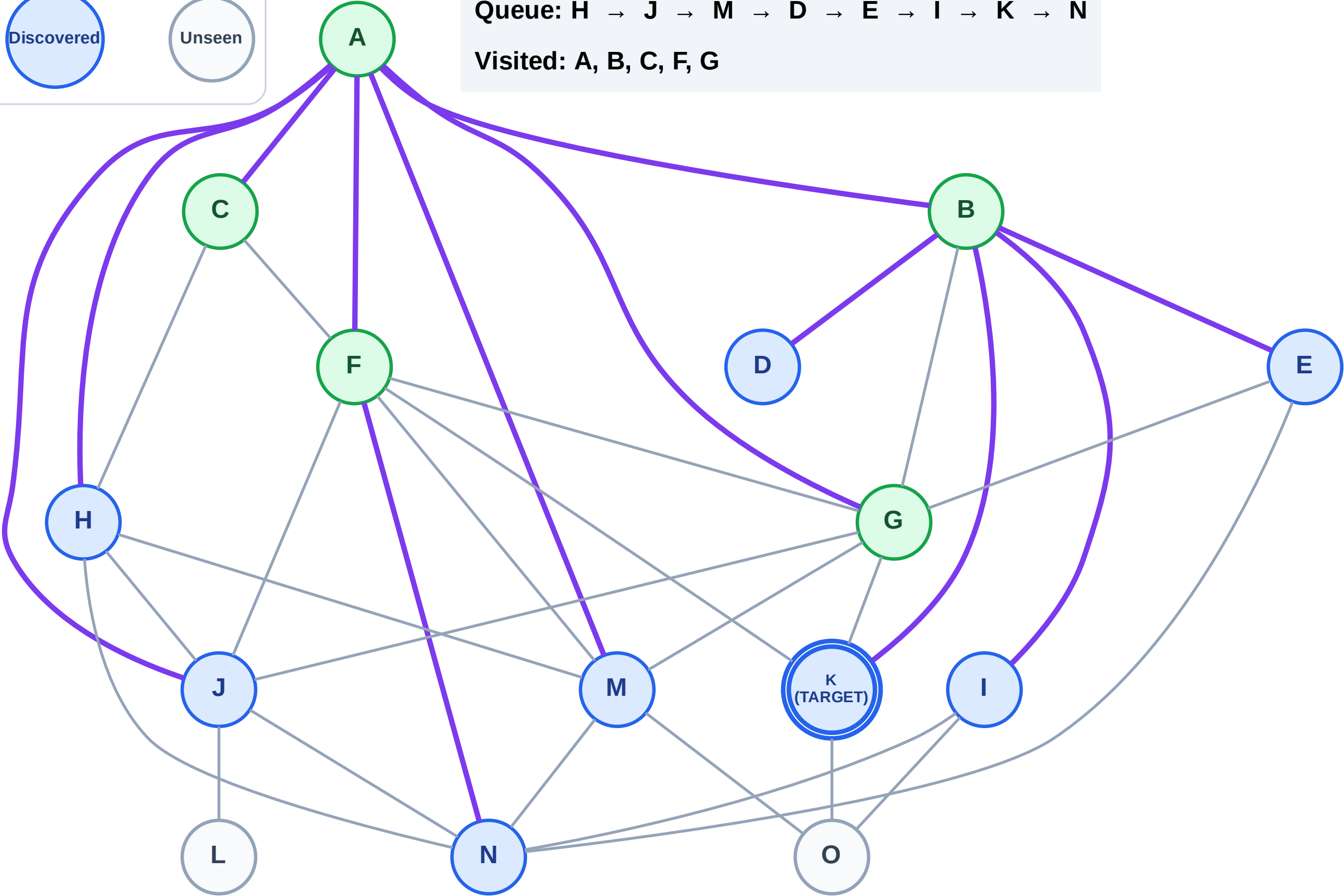
Processing

Discovered

Unseen

Queue: H → J → M → D → E → I → K → N

Visited: A, B, C, F, G



BFS Traversal

Step 12: Process node H

Legend

Complete

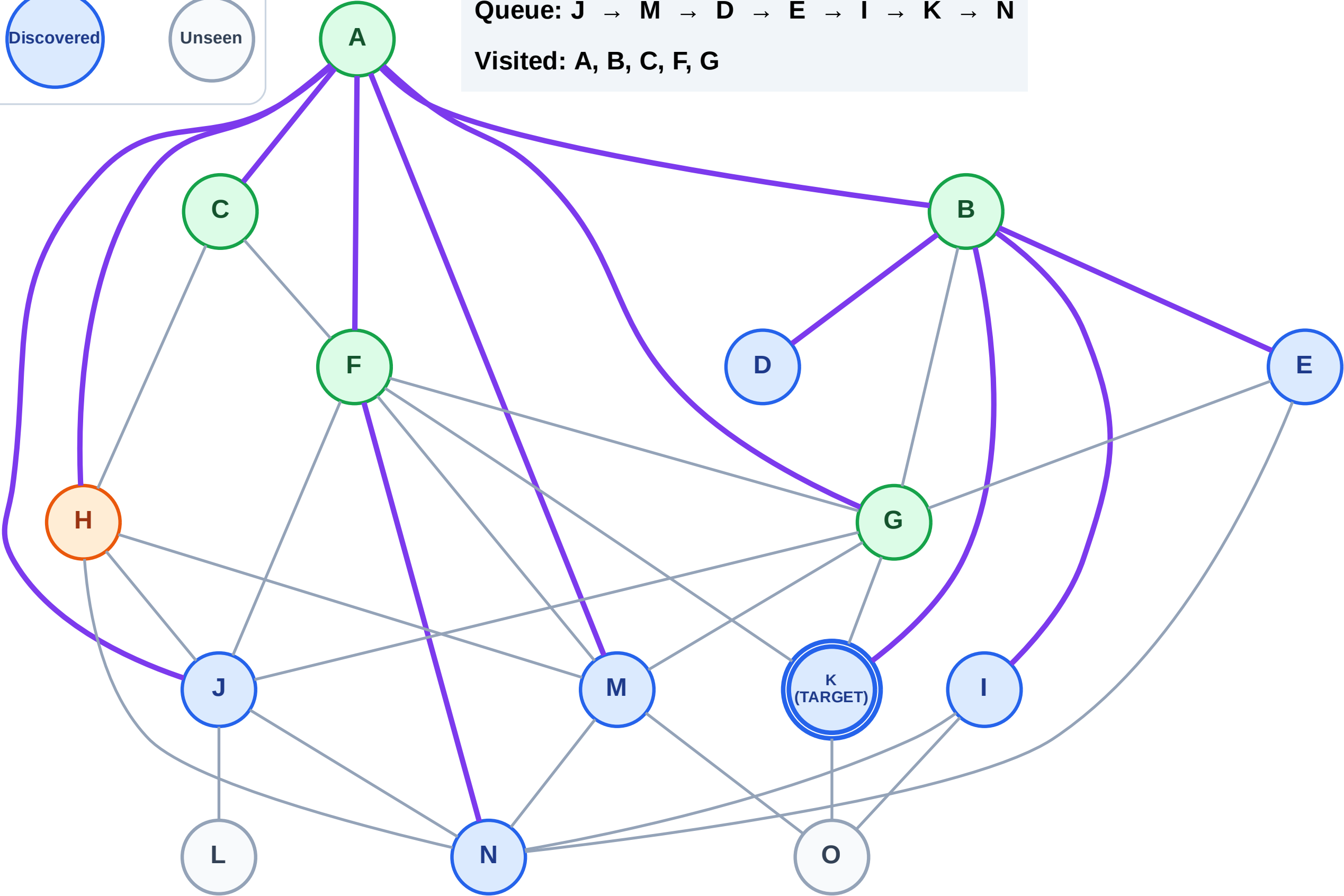
Processing

Discovered

Unseen

Queue: J → M → D → E → I → K → N

Visited: A, B, C, F, G



BFS Traversal

Step 13: No new neighbors from H

Legend

Complete

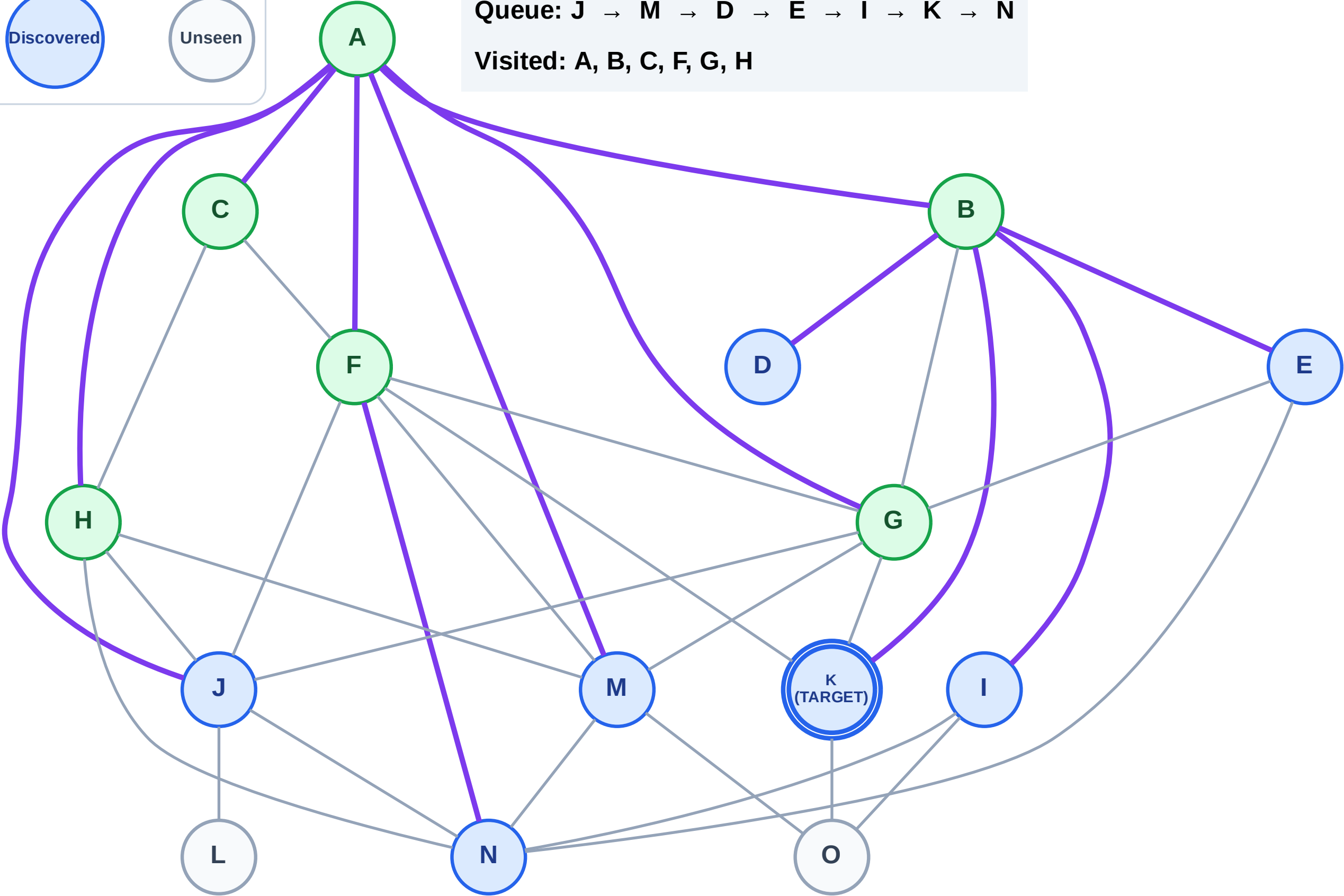
Processing

Discovered

Unseen

Queue: J → M → D → E → I → K → N

Visited: A, B, C, F, G, H



BFS Traversal

Step 14: Process node J

Legend

Complete

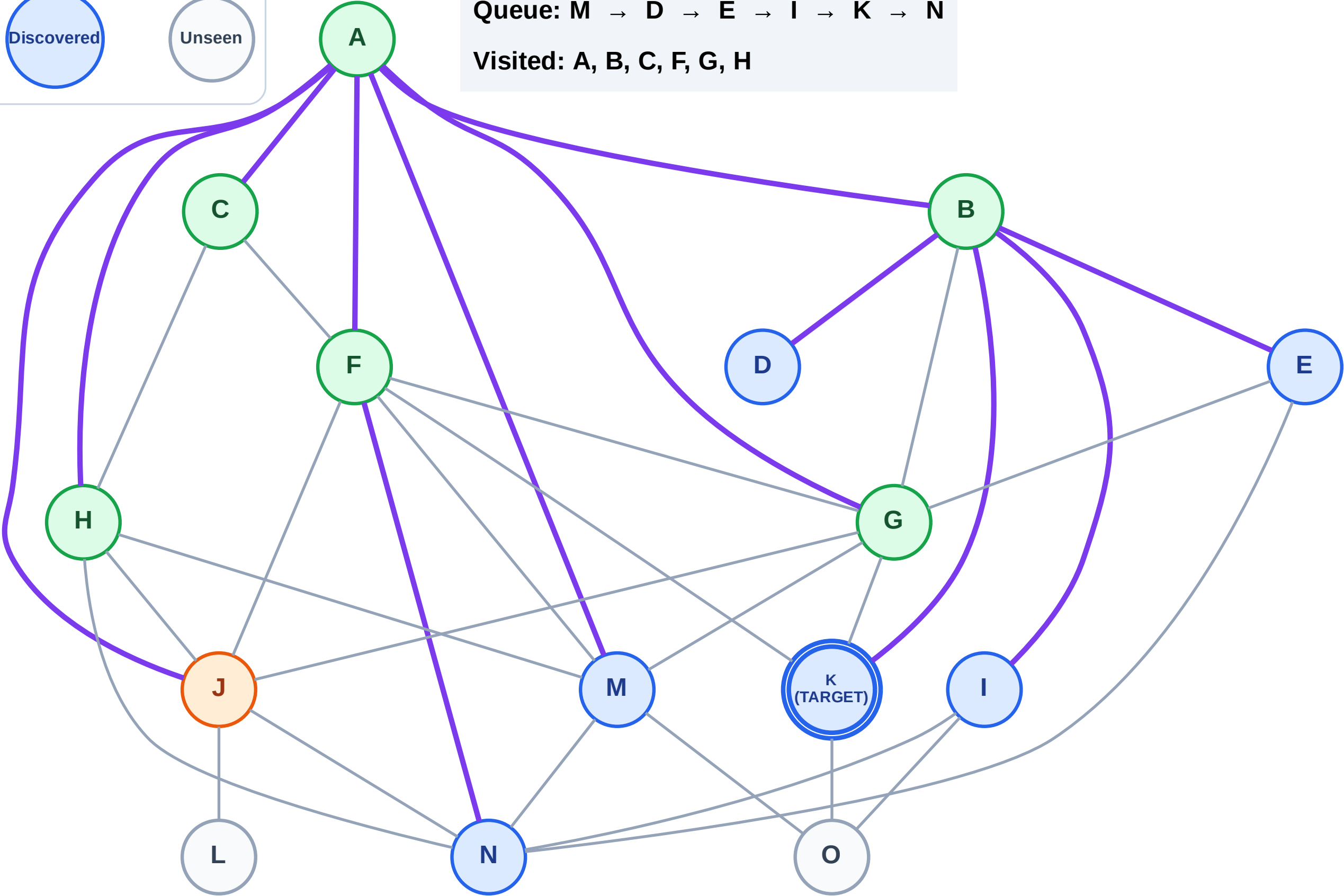
Processing

Discovered

Unseen

Queue: M → D → E → I → K → N

Visited: A, B, C, F, G, H



BFS Traversal

Step 15: Discover L from J

Legend

Complete

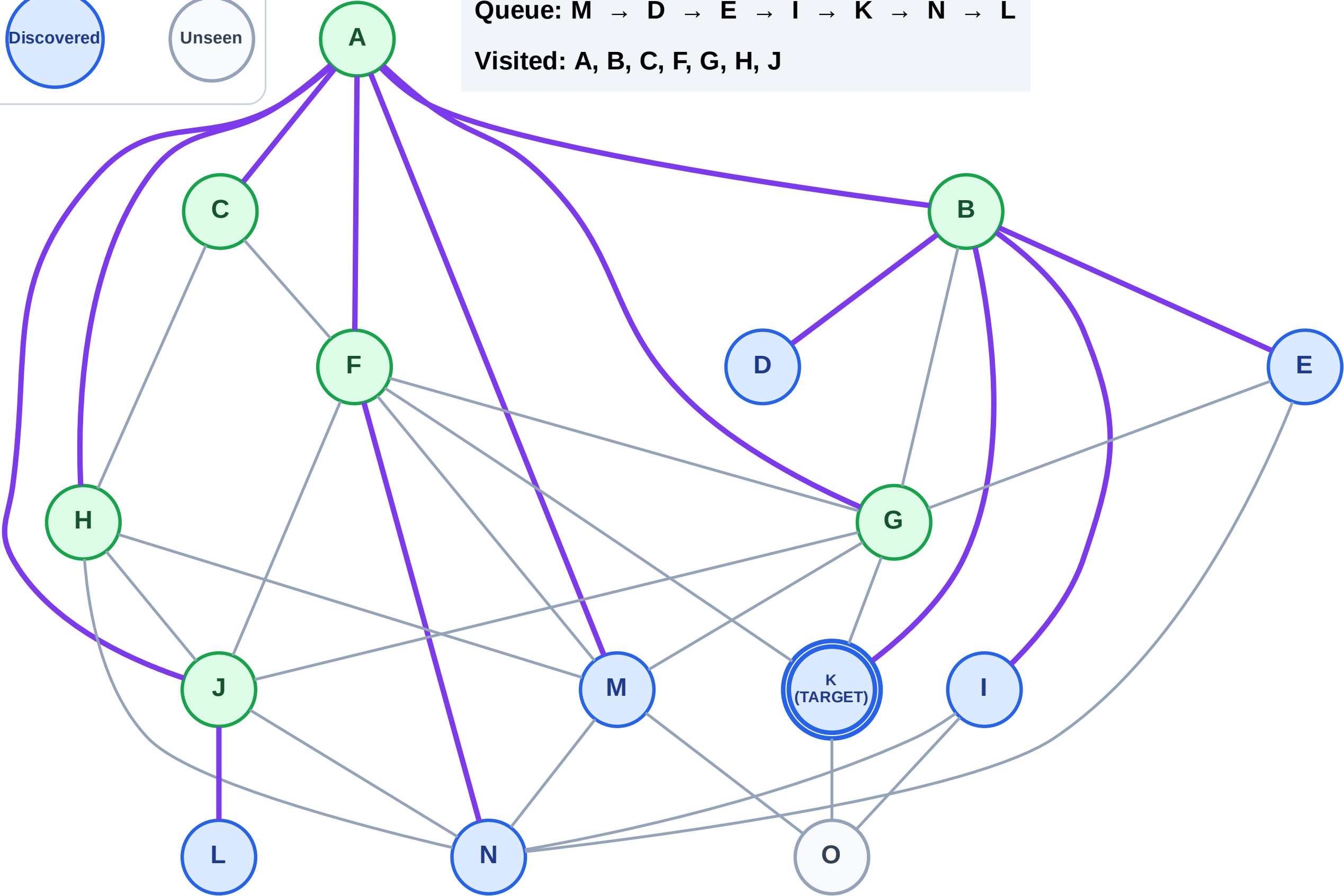
Processing

Discovered

Unseen

Queue: M → D → E → I → K → N → L

Visited: A, B, C, F, G, H, J



BFS Traversal

Step 16: Process node M

Legend

Complete

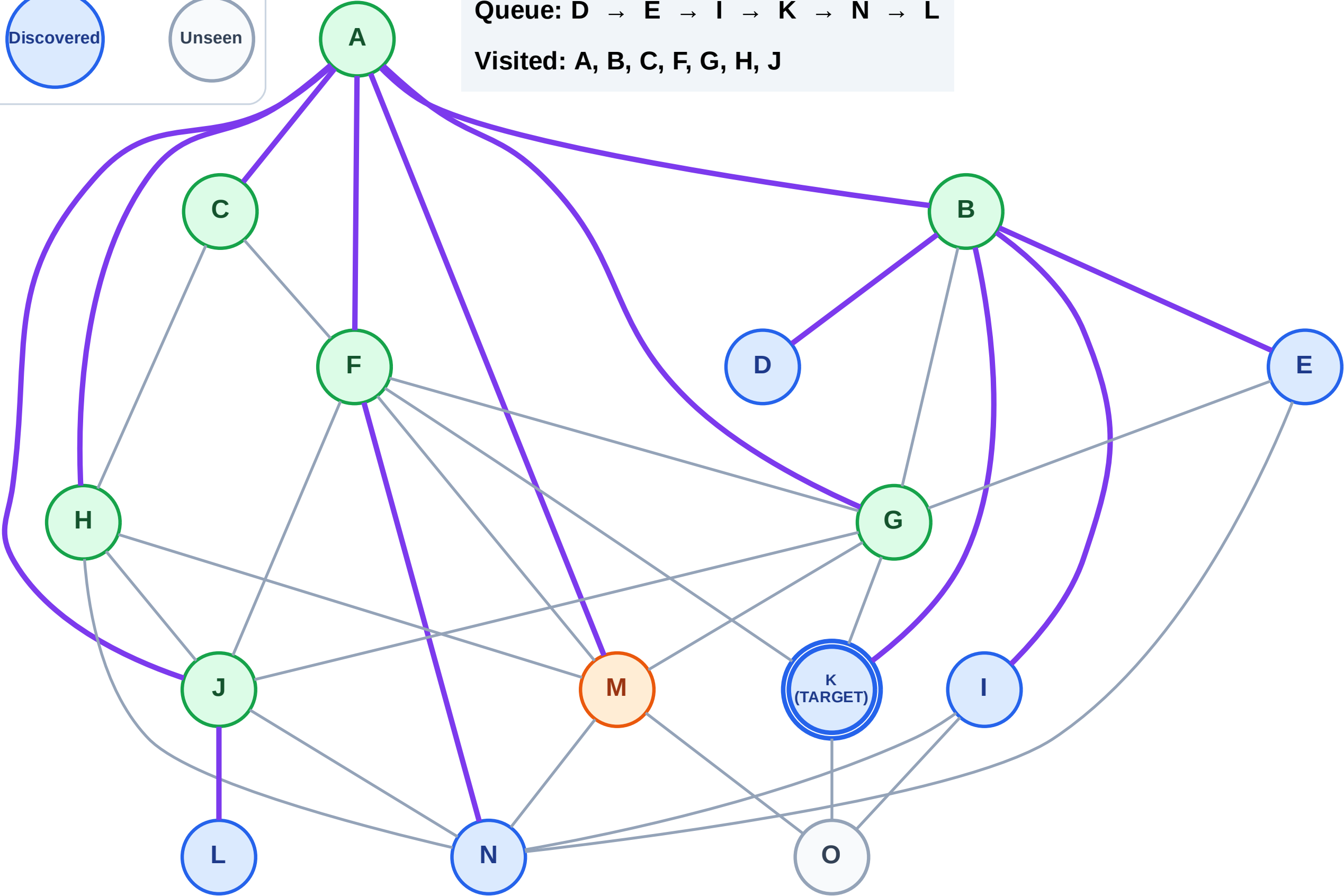
Processing

Discovered

Unseen

Queue: D → E → I → K → N → L

Visited: A, B, C, F, G, H, J



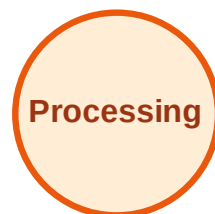
BFS Traversal

Step 17: Discover O from M

Legend



Complete



Processing



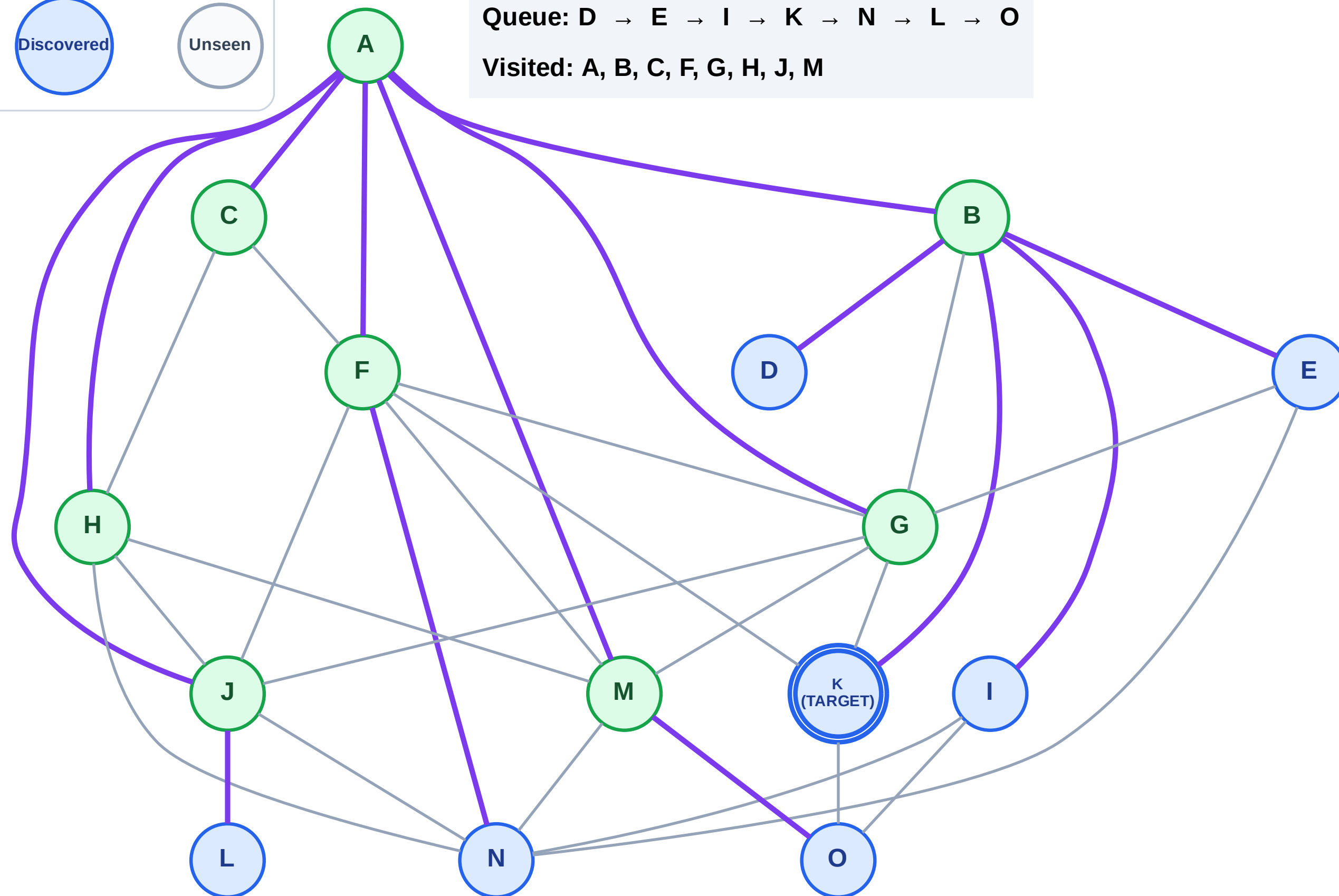
Discovered



Unseen

Queue: D → E → I → K → N → L → O

Visited: A, B, C, F, G, H, J, M



BFS Traversal

Step 18: Process node D

Legend

Complete

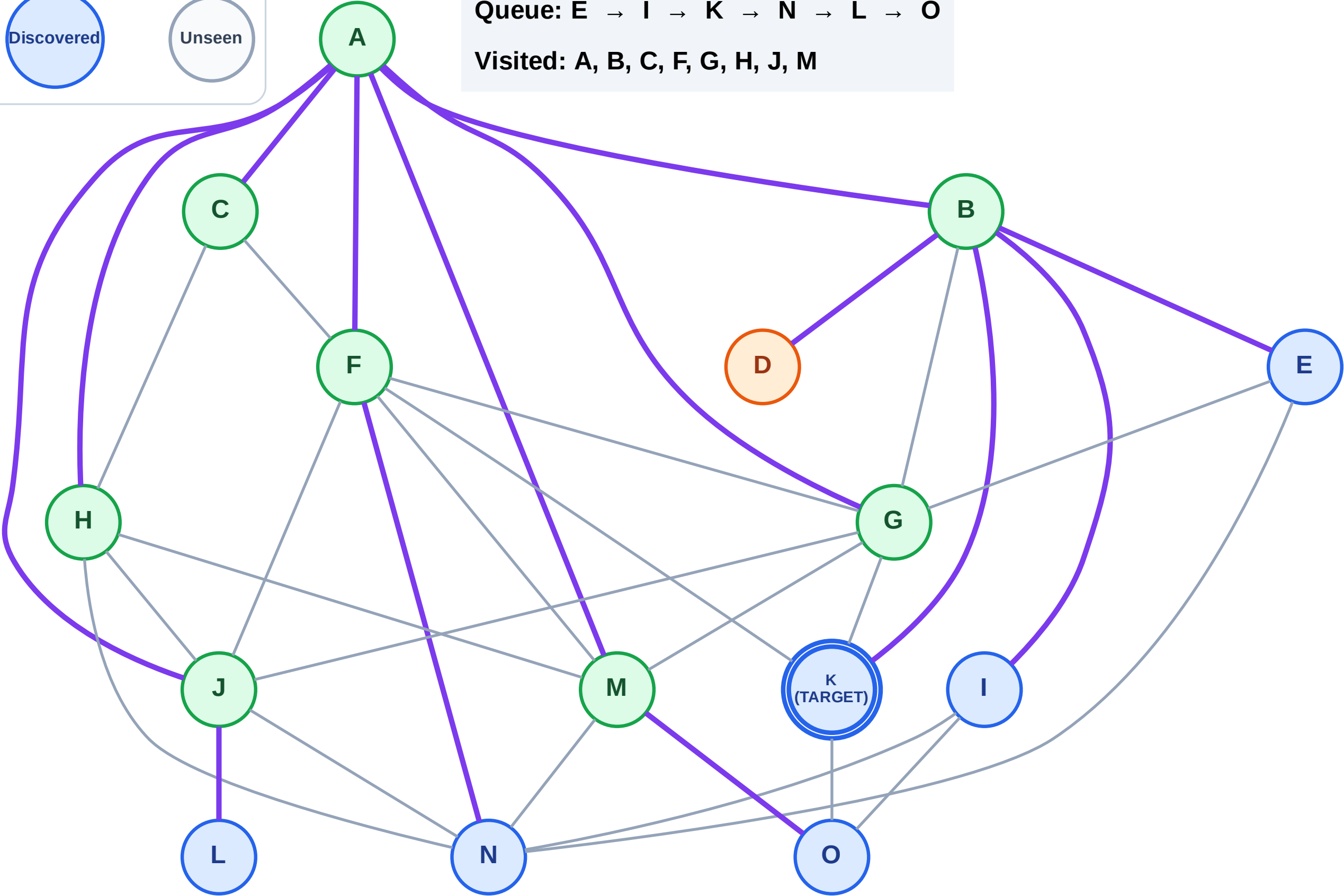
Processing

Discovered

Unseen

Queue: E → I → K → N → L → O

Visited: A, B, C, F, G, H, J, M



BFS Traversal

Step 19: No new neighbors from D

Legend

Complete

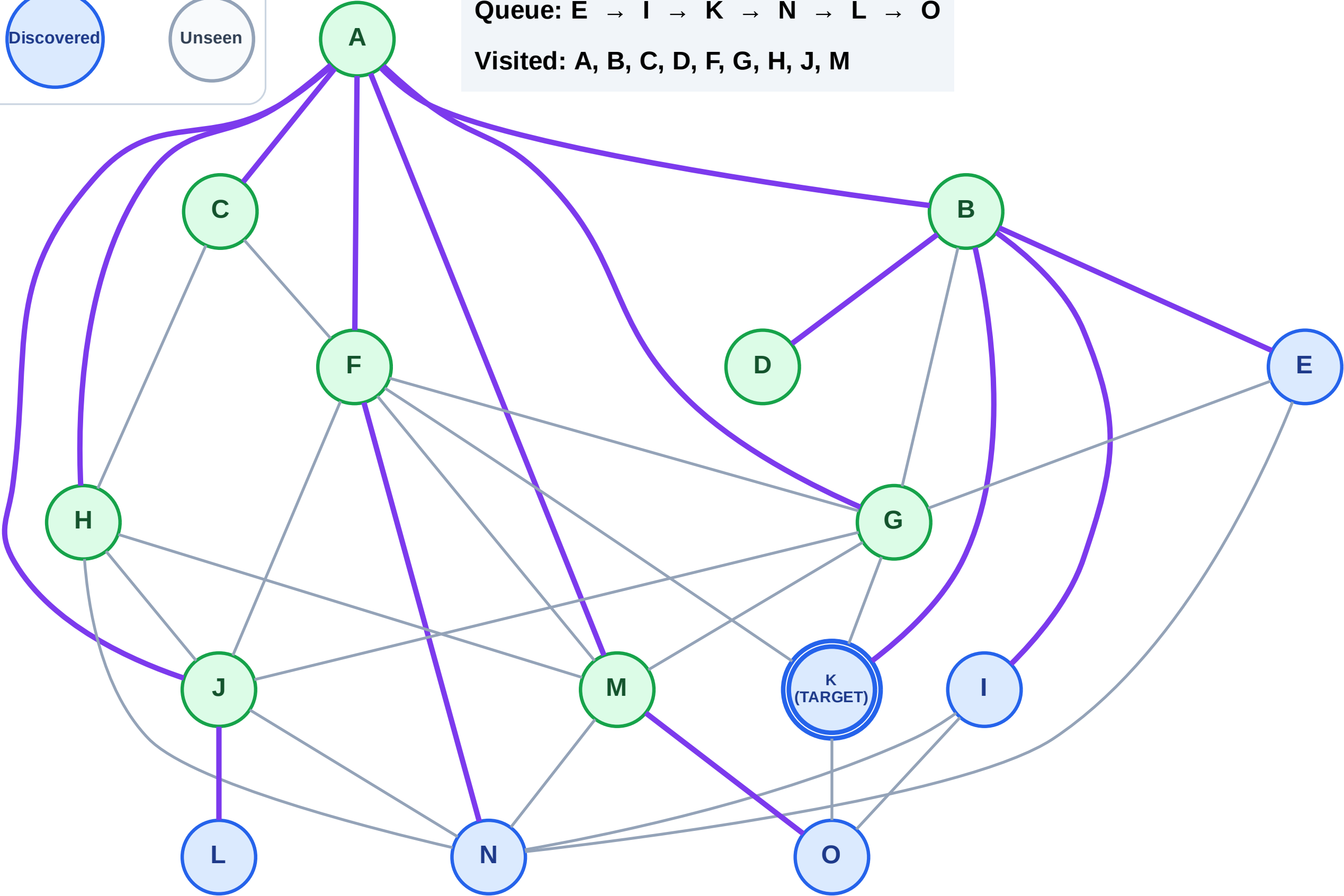
Processing

Discovered

Unseen

Queue: E → I → K → N → L → O

Visited: A, B, C, D, F, G, H, J, M



BFS Traversal

Step 20: Process node E

Legend

Complete

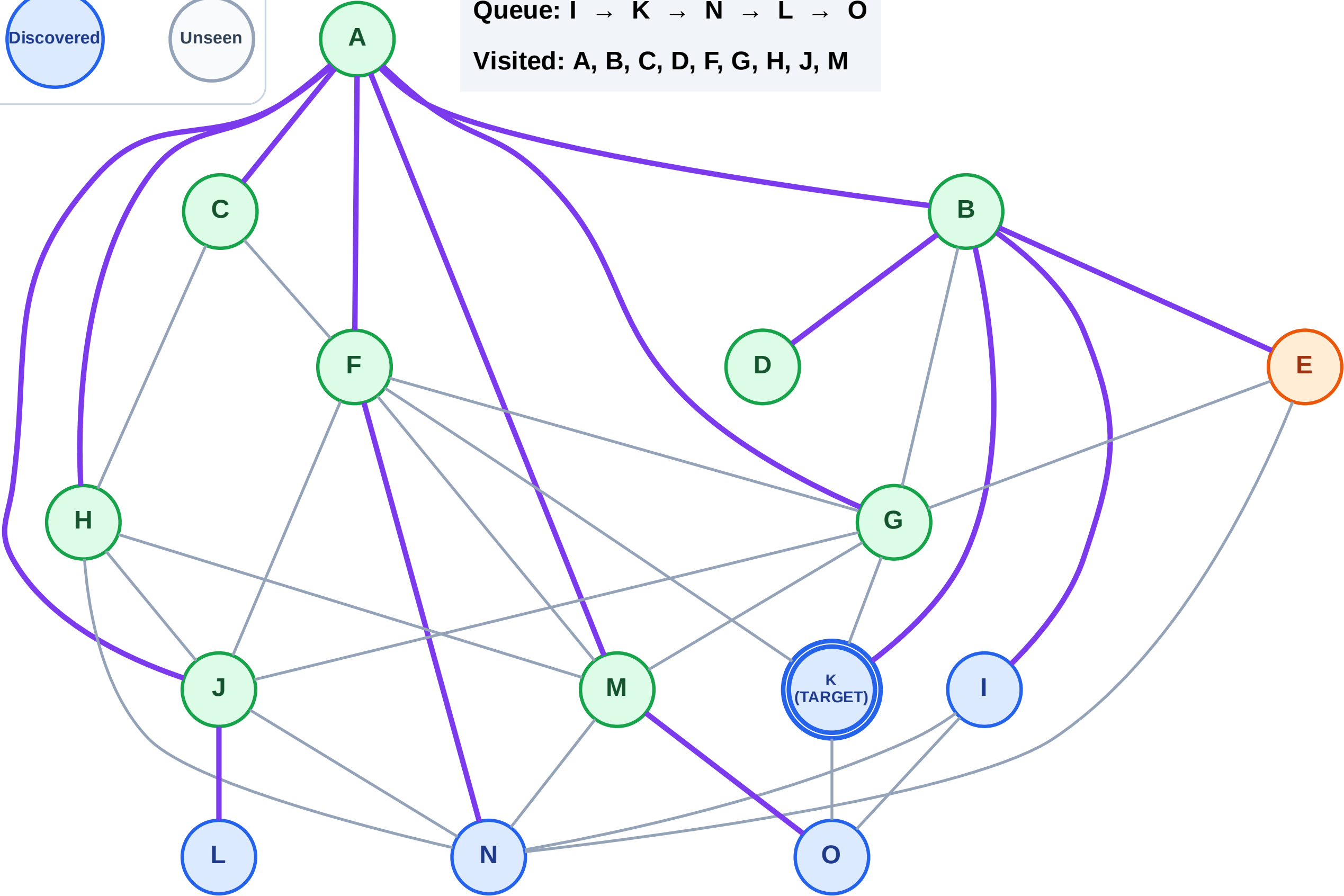
Processing

Discovered

Unseen

Queue: I → K → N → L → O

Visited: A, B, C, D, F, G, H, J, M



BFS Traversal

Step 21: No new neighbors from E

Legend

Complete

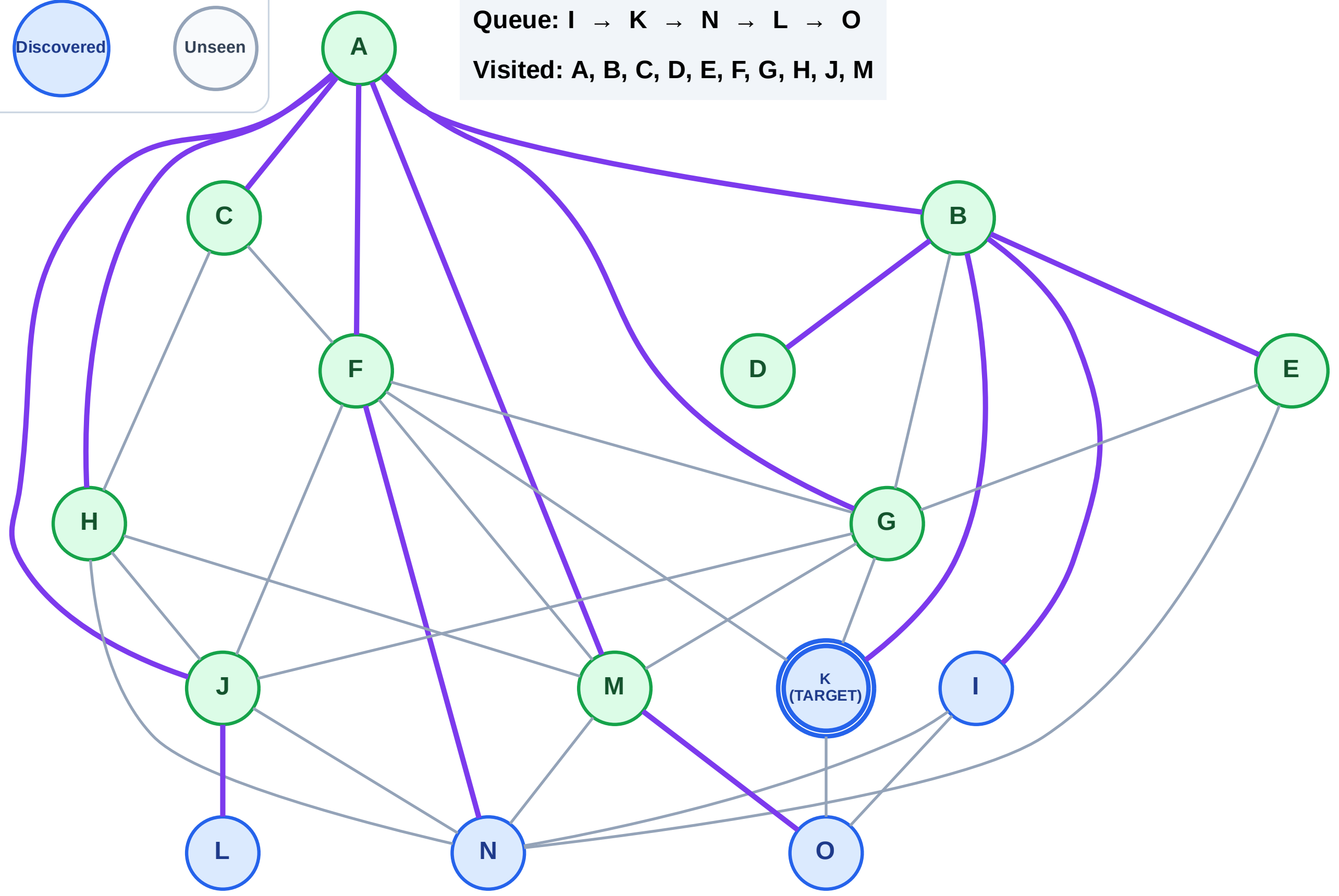
Processing

Discovered

Unseen

Queue: I → K → N → L → O

Visited: A, B, C, D, E, F, G, H, J, M



BFS Traversal

Step 22: Process node I

Legend

Complete

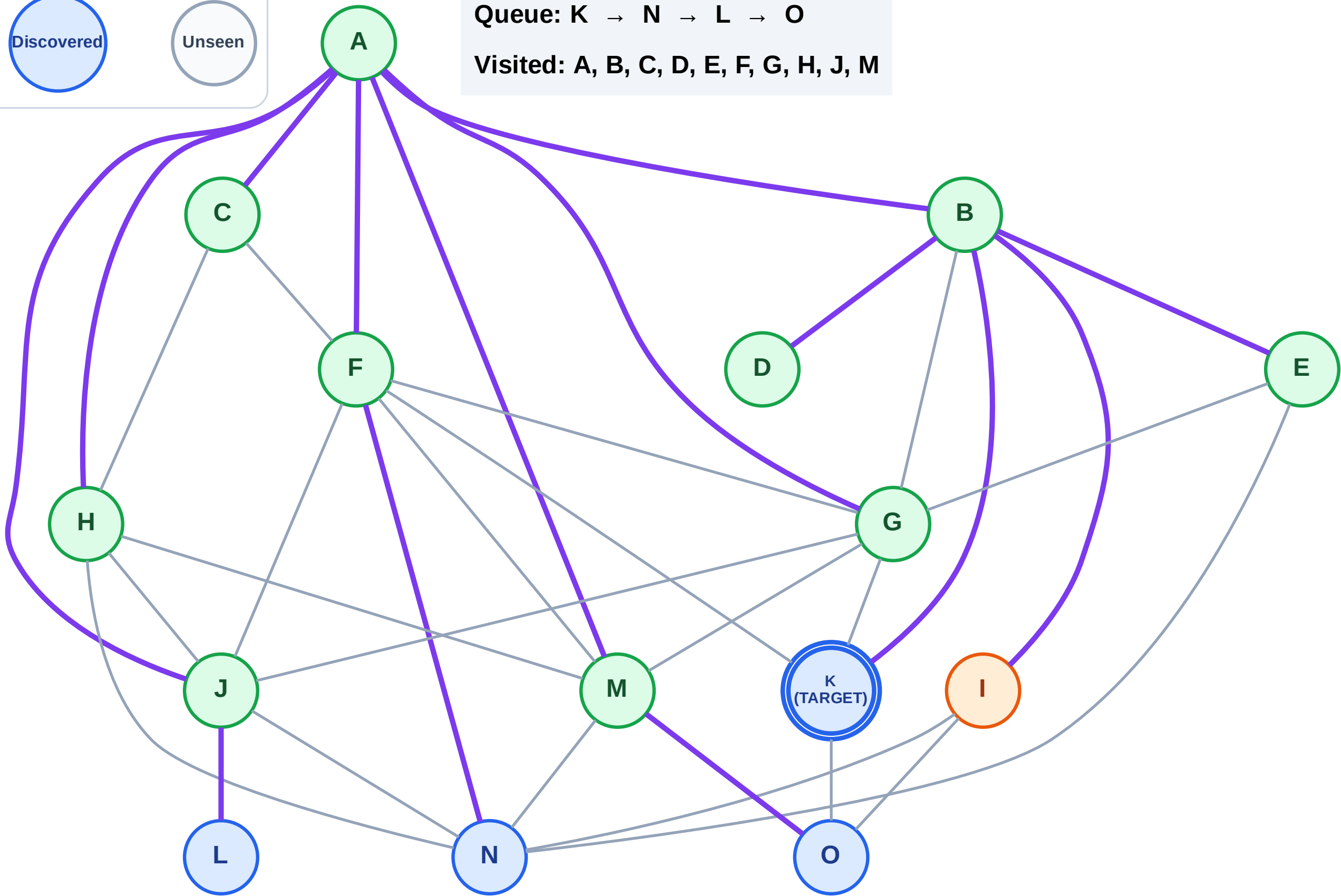
Processing

Discovered

Unseen

Queue: K → N → L → O

Visited: A, B, C, D, E, F, G, H, J, M



BFS Traversal

Step 23: No new neighbors from I

Legend

Complete

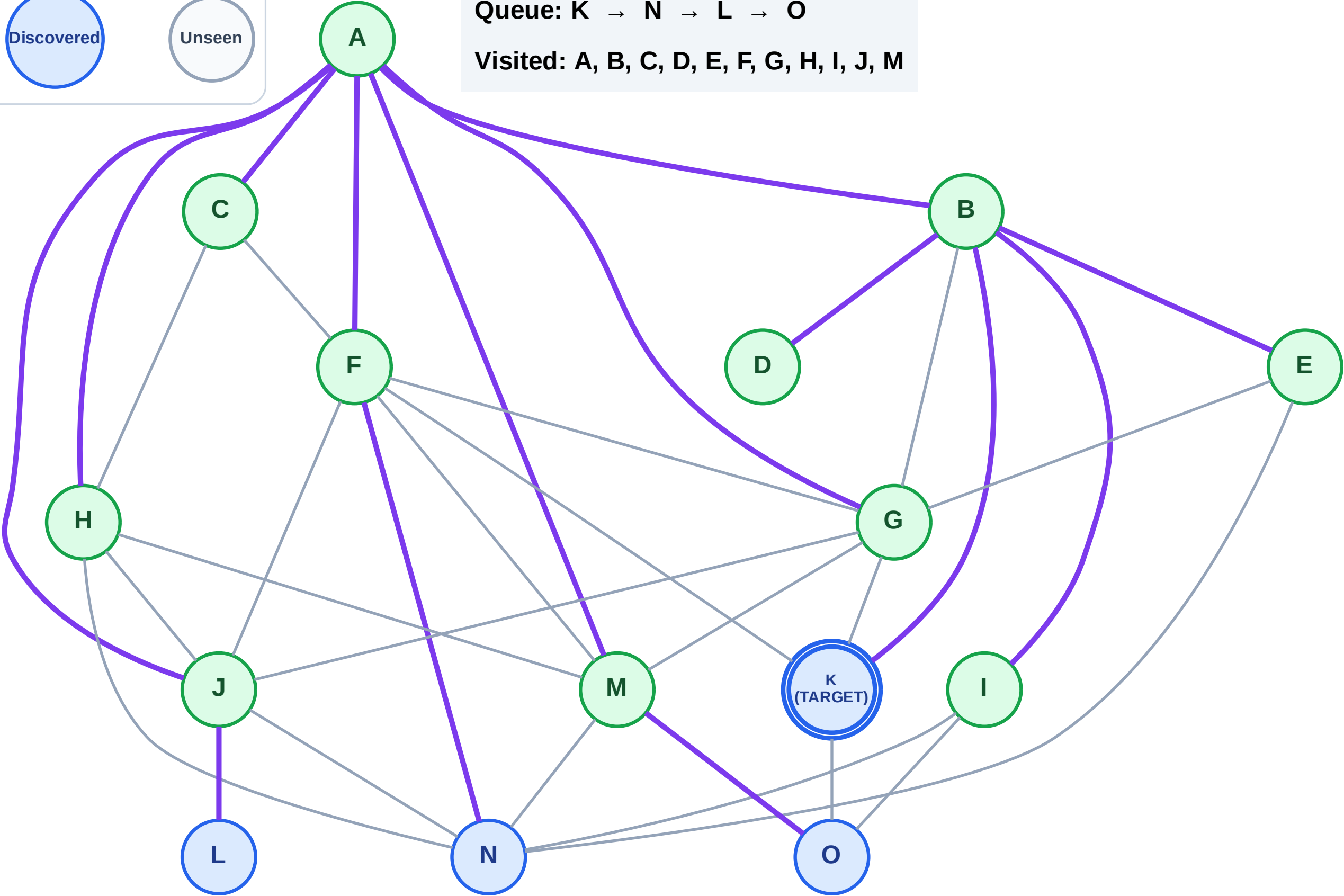
Processing

Discovered

Unseen

Queue: K → N → L → O

Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

Step 24: Process node K

Legend

Complete

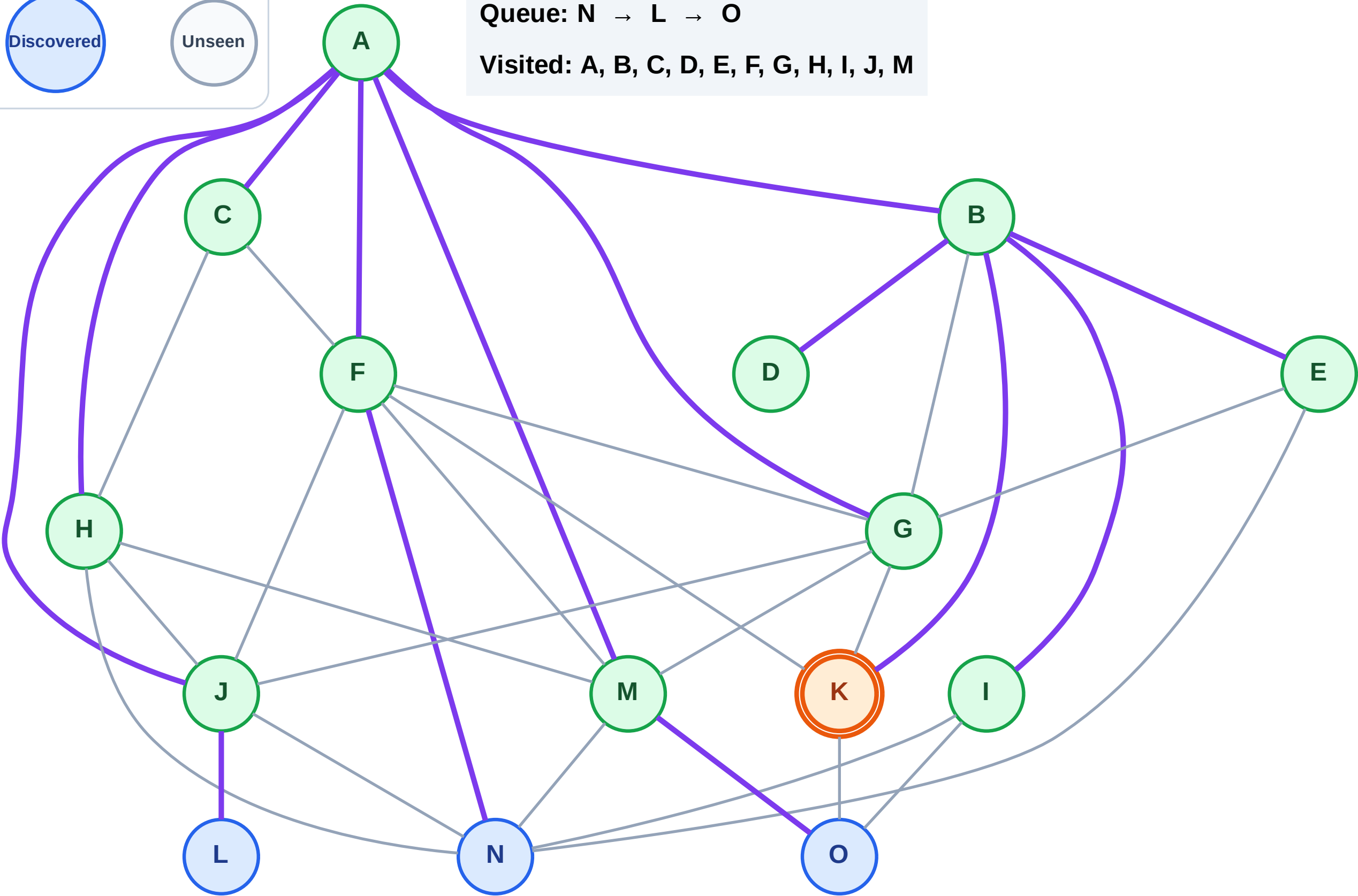
Processing

Discovered

Unseen

Queue: N → L → O

Visited: A, B, C, D, E, F, G, H, I, J, M



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

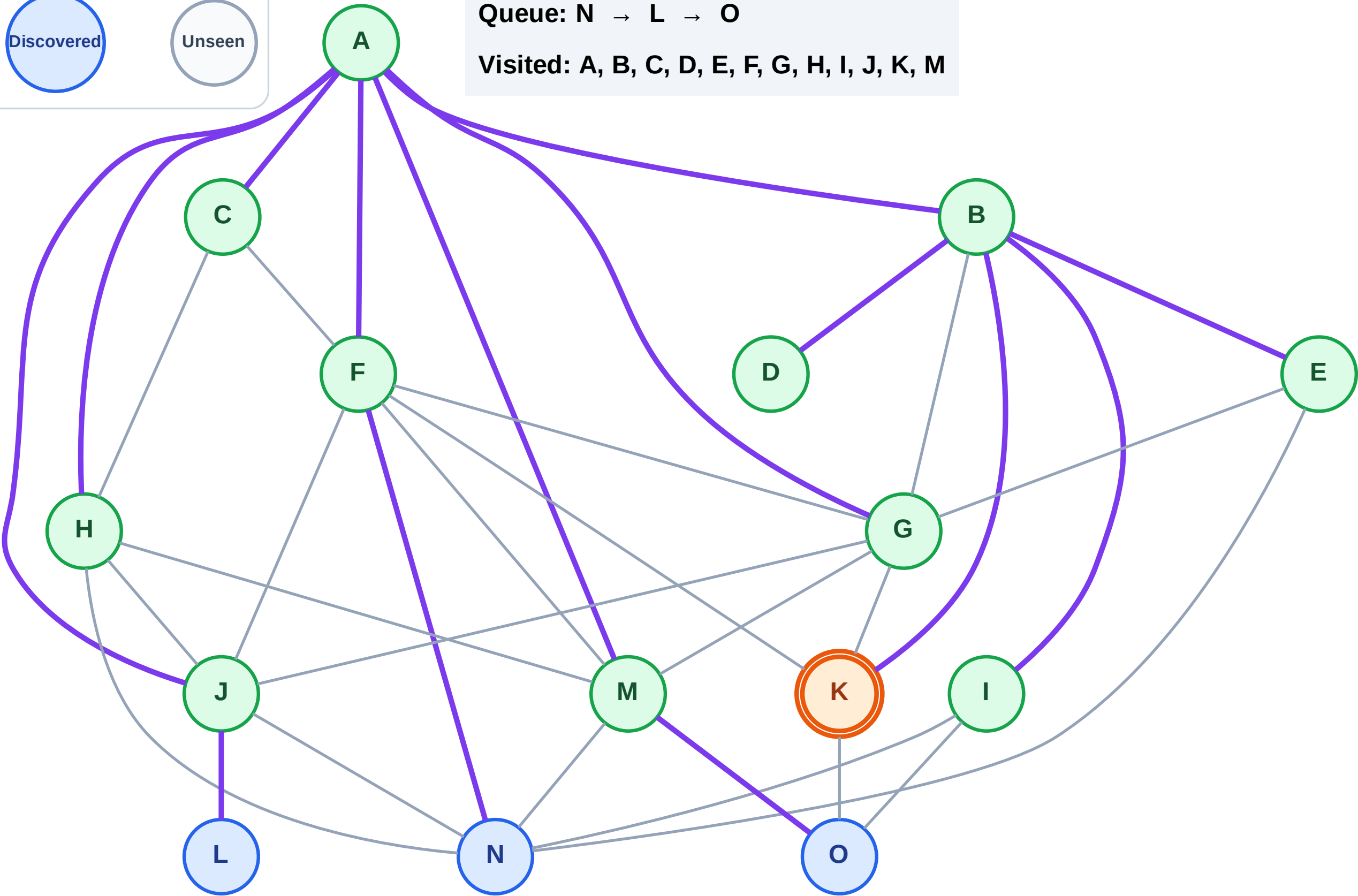
Processing

Discovered

Unseen

Queue: N → L → O

Visited: A, B, C, D, E, F, G, H, I, J, K, M



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

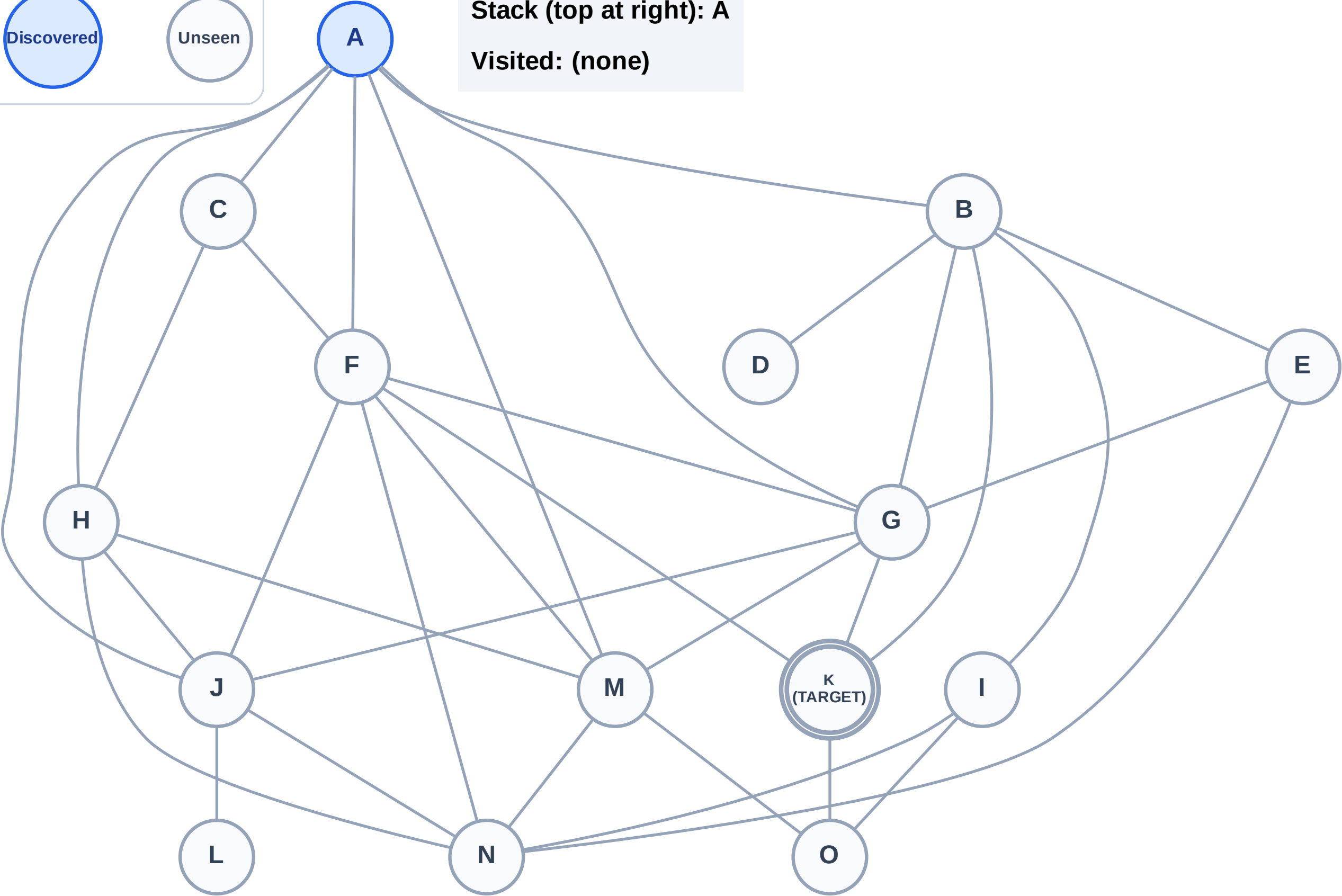
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

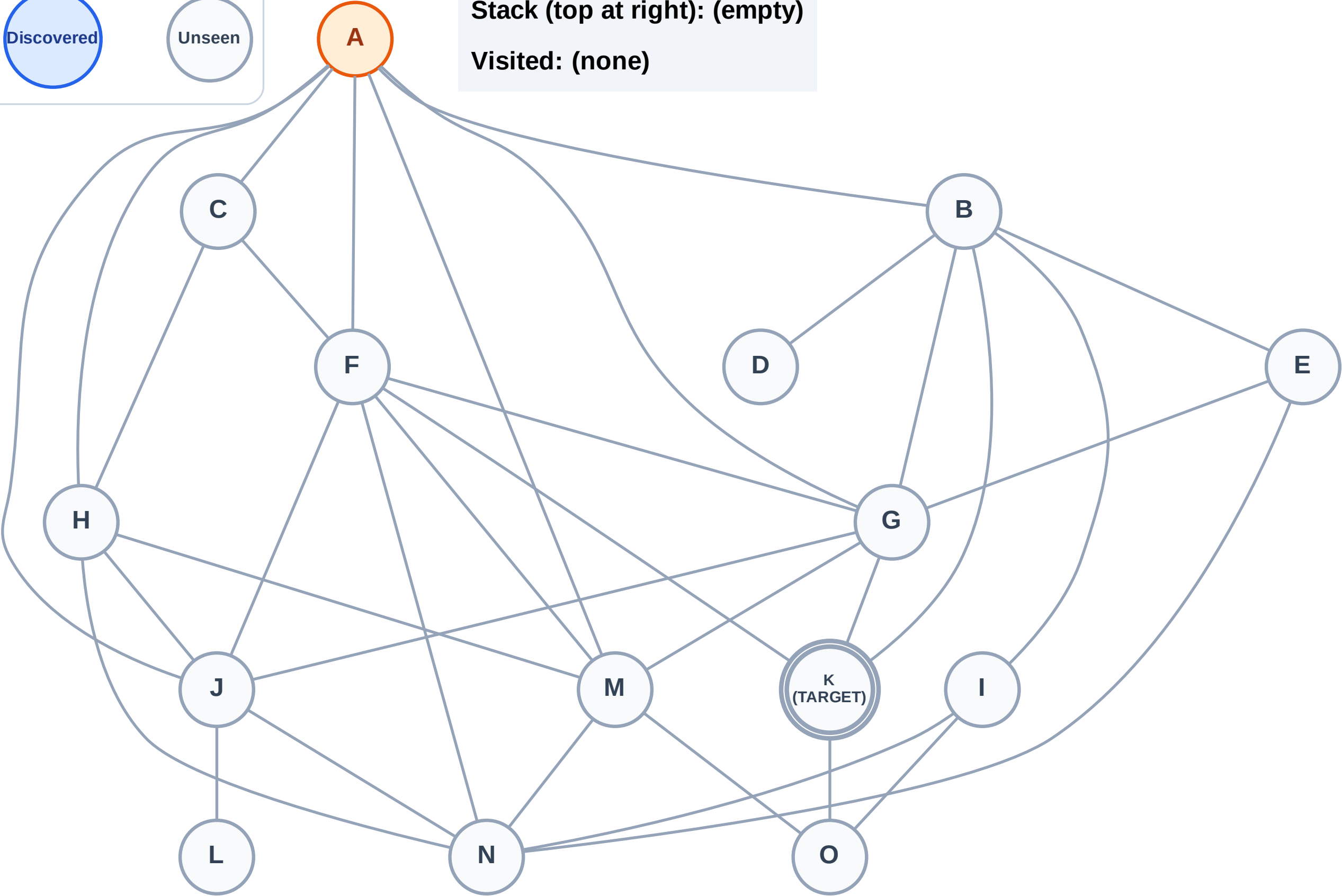
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover M, J, H, G, F, C, B from A

Legend

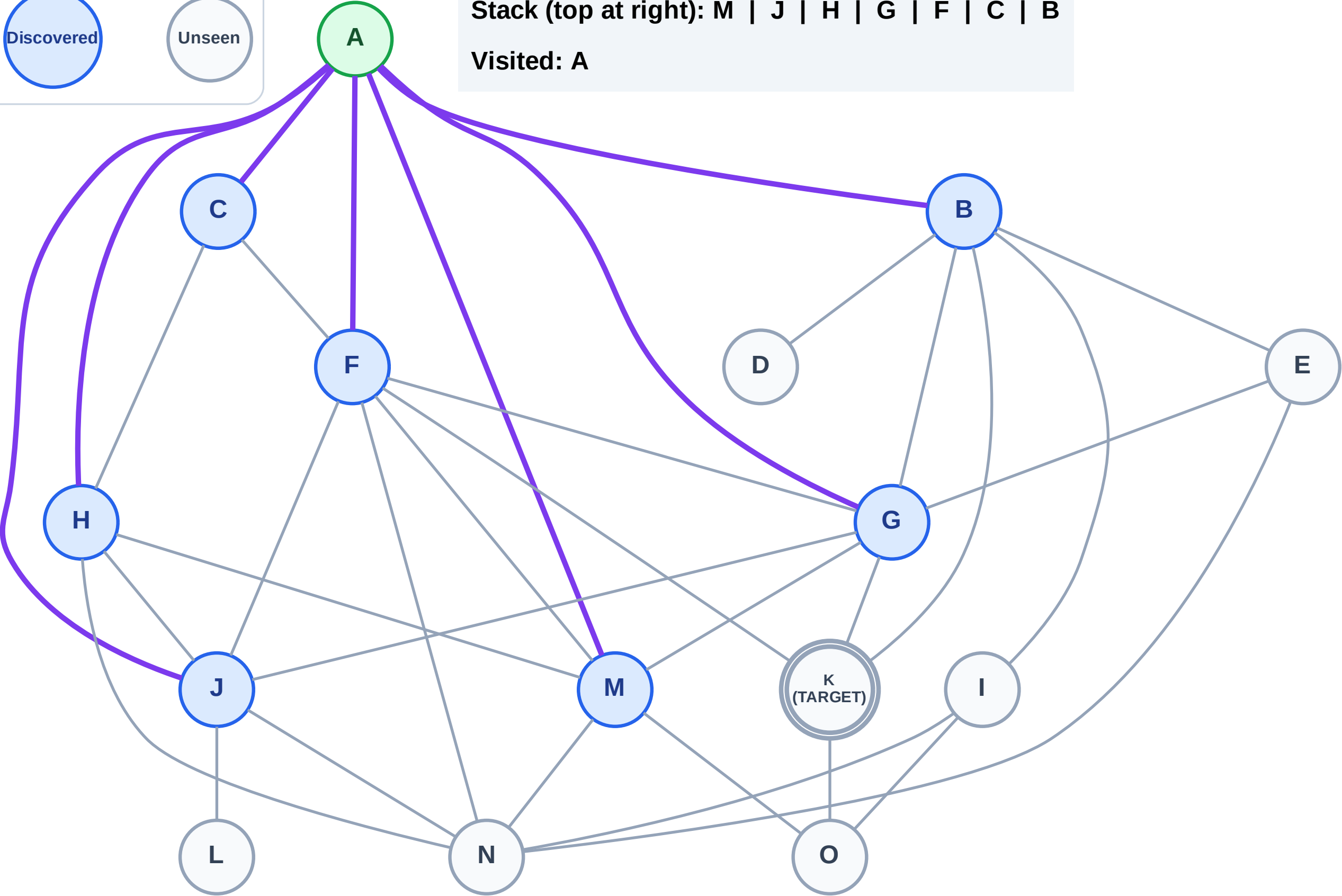
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

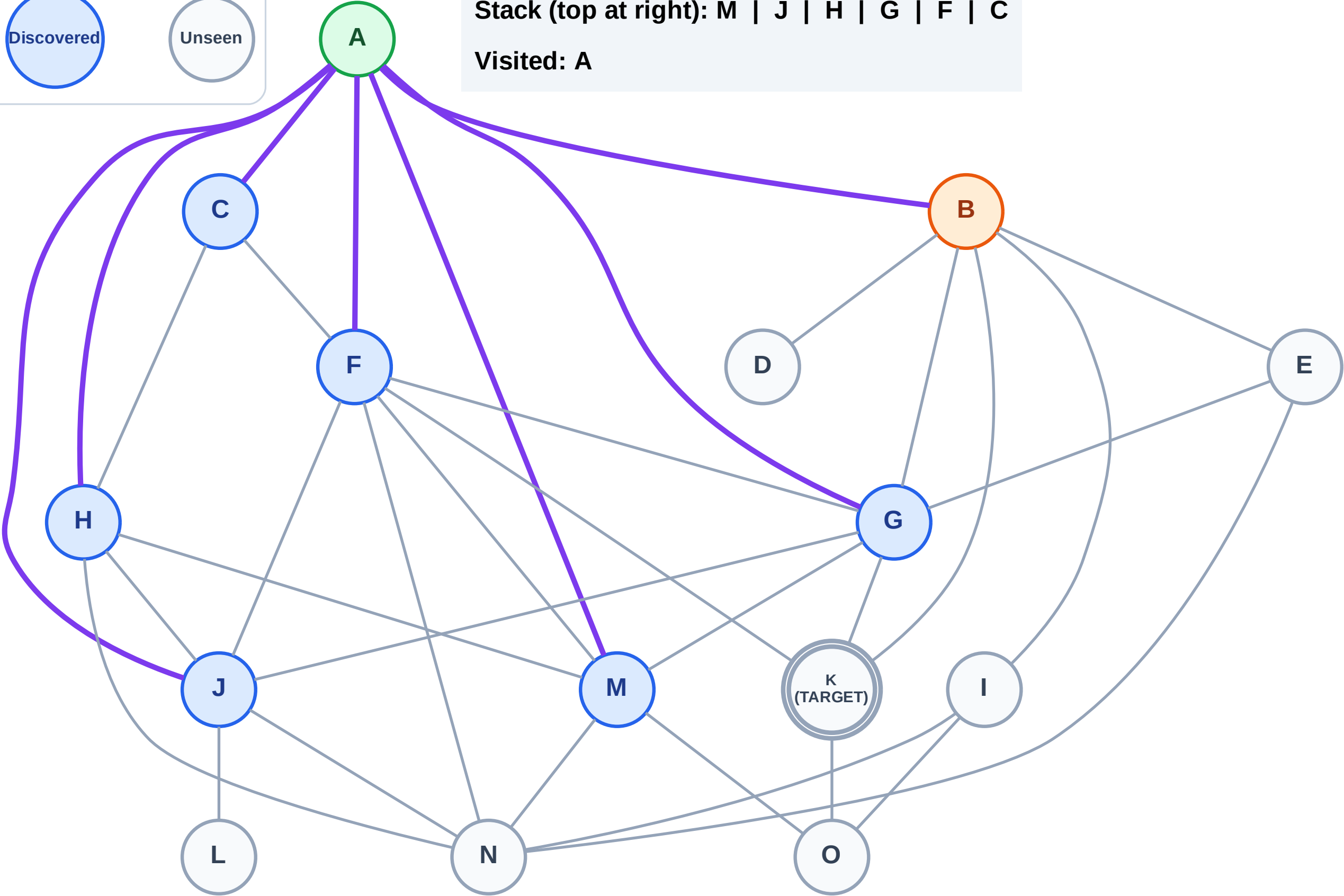
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C

Visited: A



DFS Traversal

Step 5: Discover K, I, E, D from B

Legend

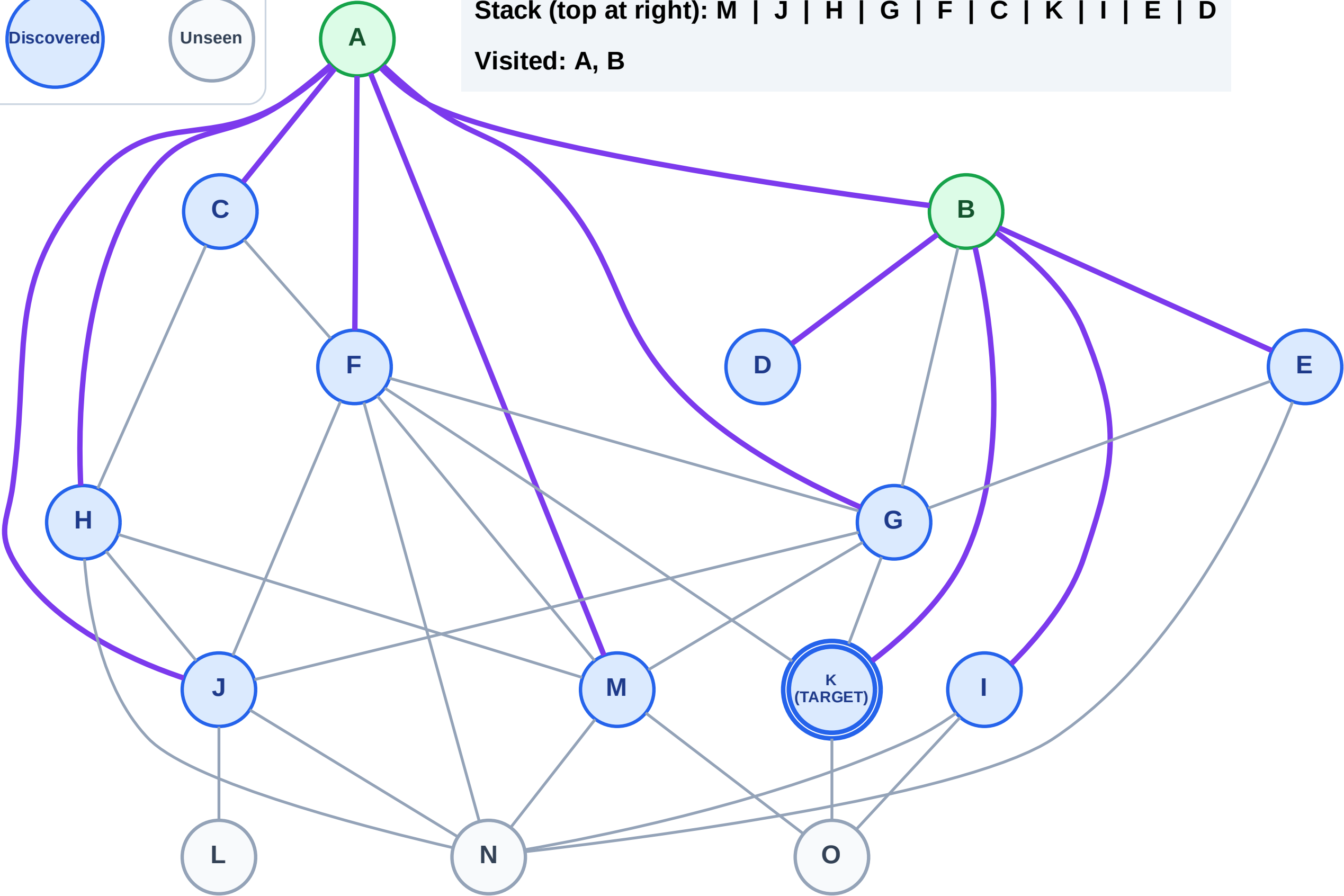
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K | I | E | D
Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

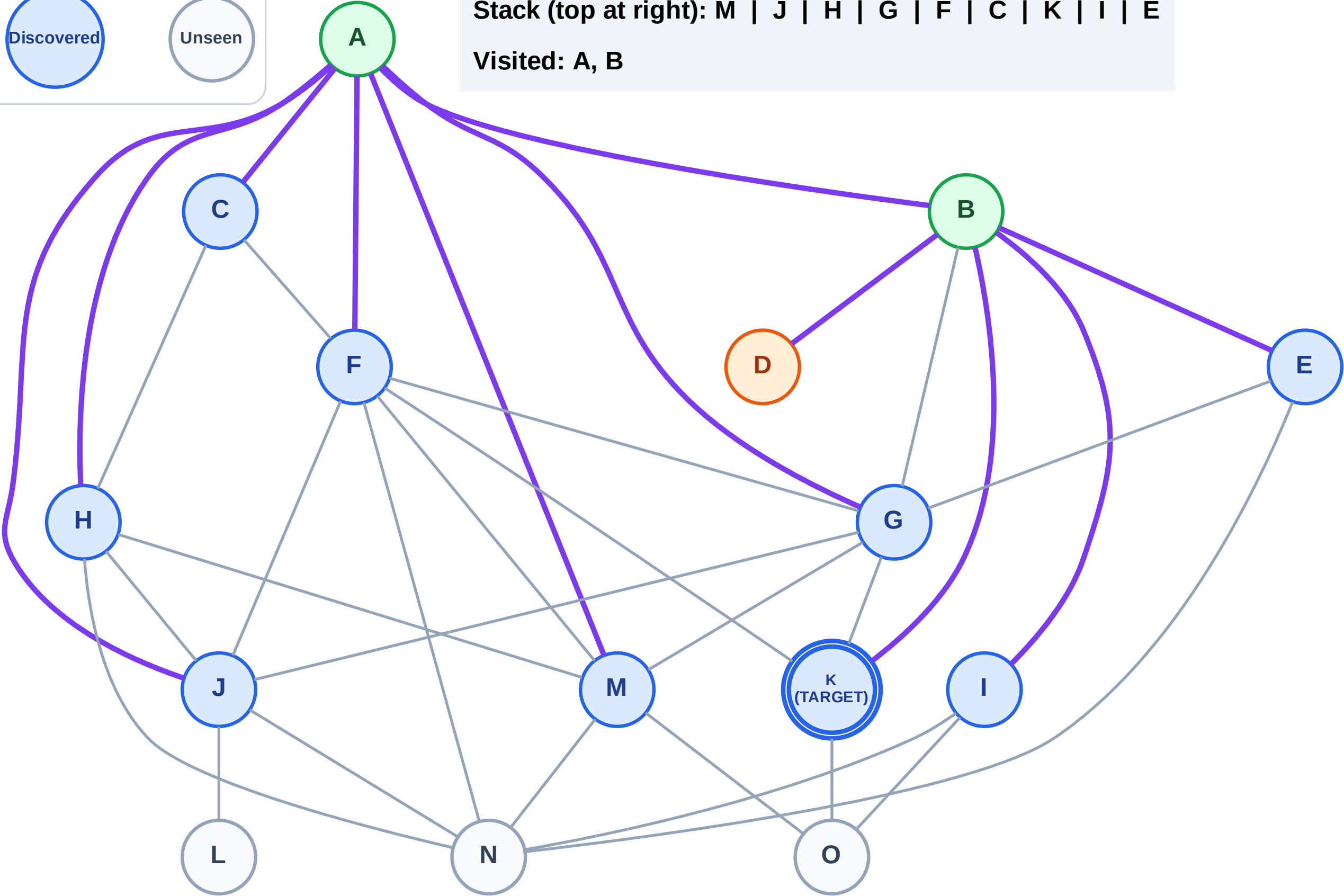
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K | I | E

Visited: A, B



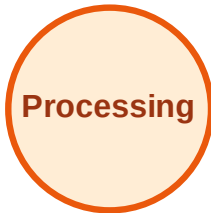
Step 7: No new neighbors from D

Stack (top at right): M | J | H | G | F | C | K | I | E

Legend

Processing

Unseen



DFS Traversal

Step 8: Process node E

Legend

Complete

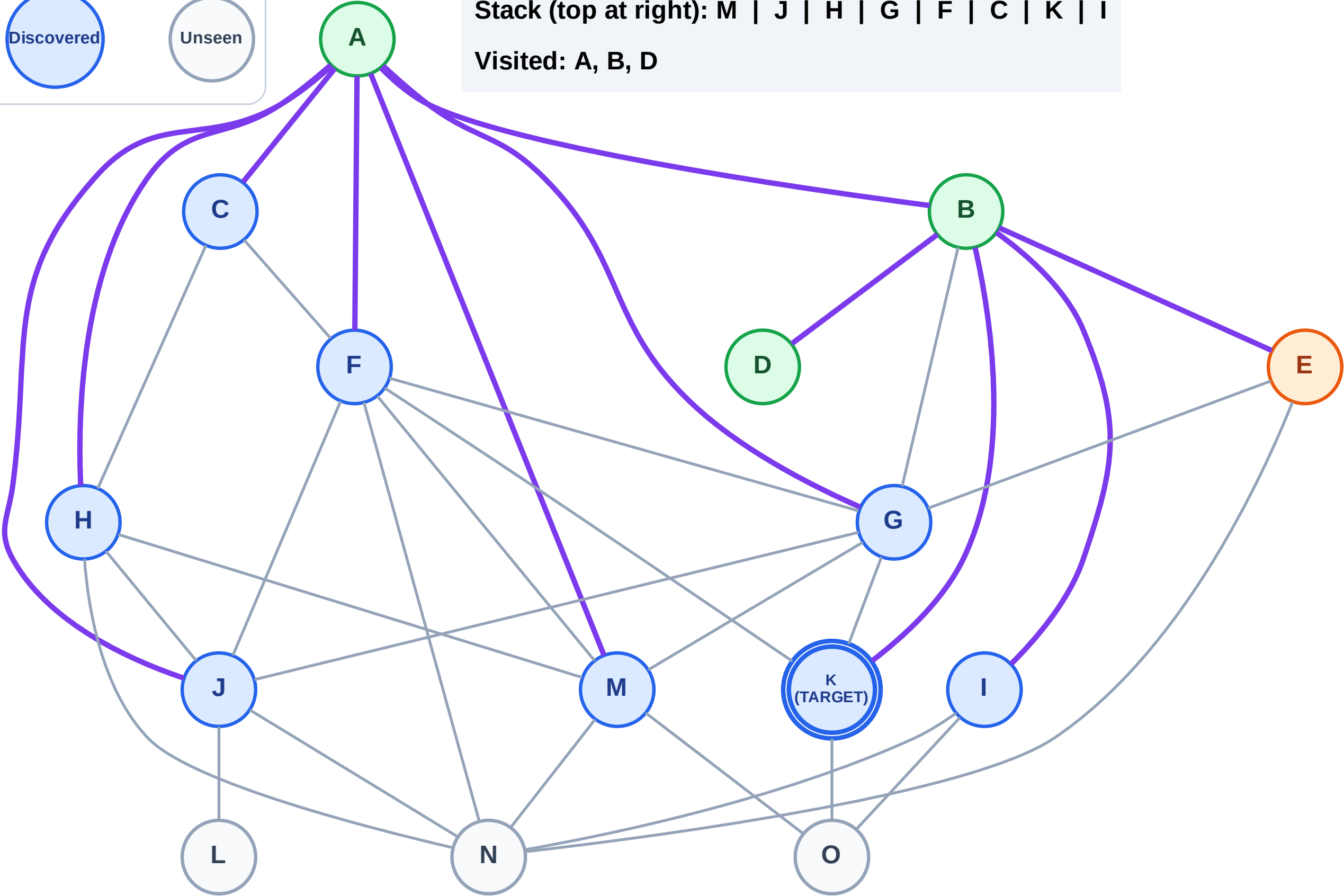
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K | I

Visited: A, B, D



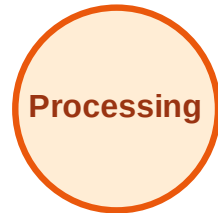
DFS Traversal

Step 9: Discover N from E

Legend



Complete



Processing

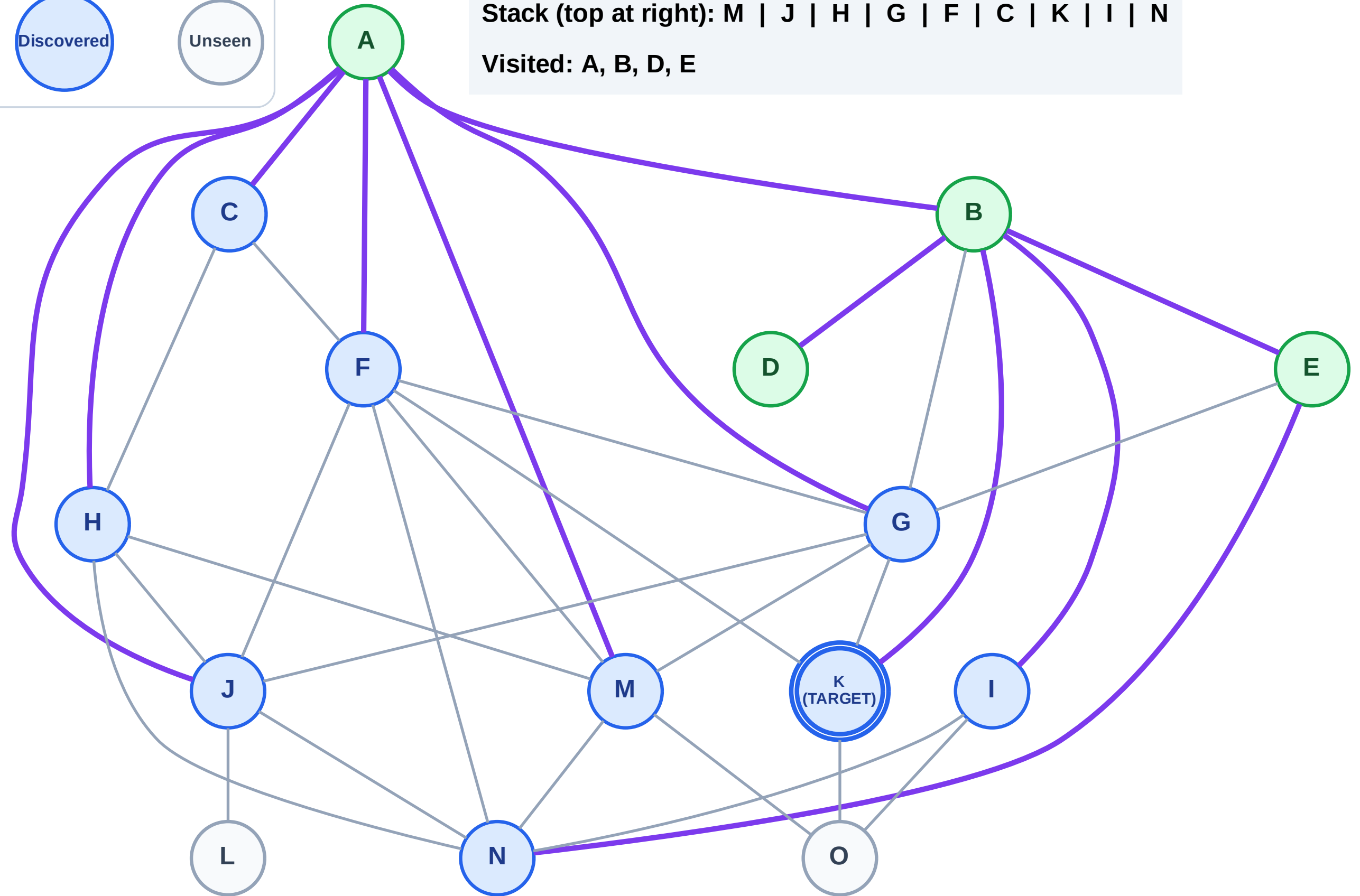


Discovered



Unseen

Stack (top at right): M | J | H | G | F | C | K | I | N
Visited: A, B, D, E



DFS Traversal

Step 10: Process node N

Legend

Complete

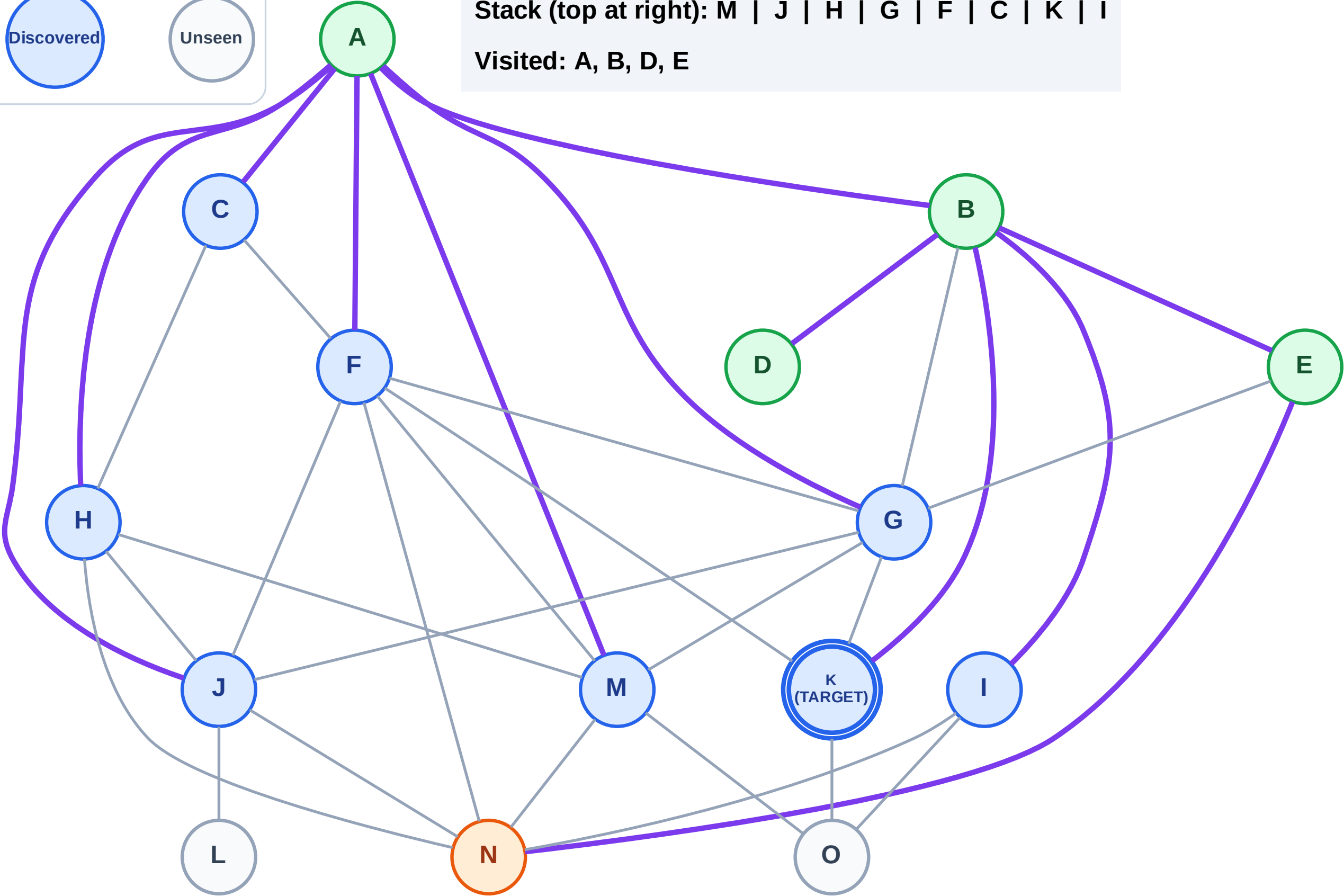
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K | I

Visited: A, B, D, E



DFS Traversal

Step 11: No new neighbors from N

Legend

Complete

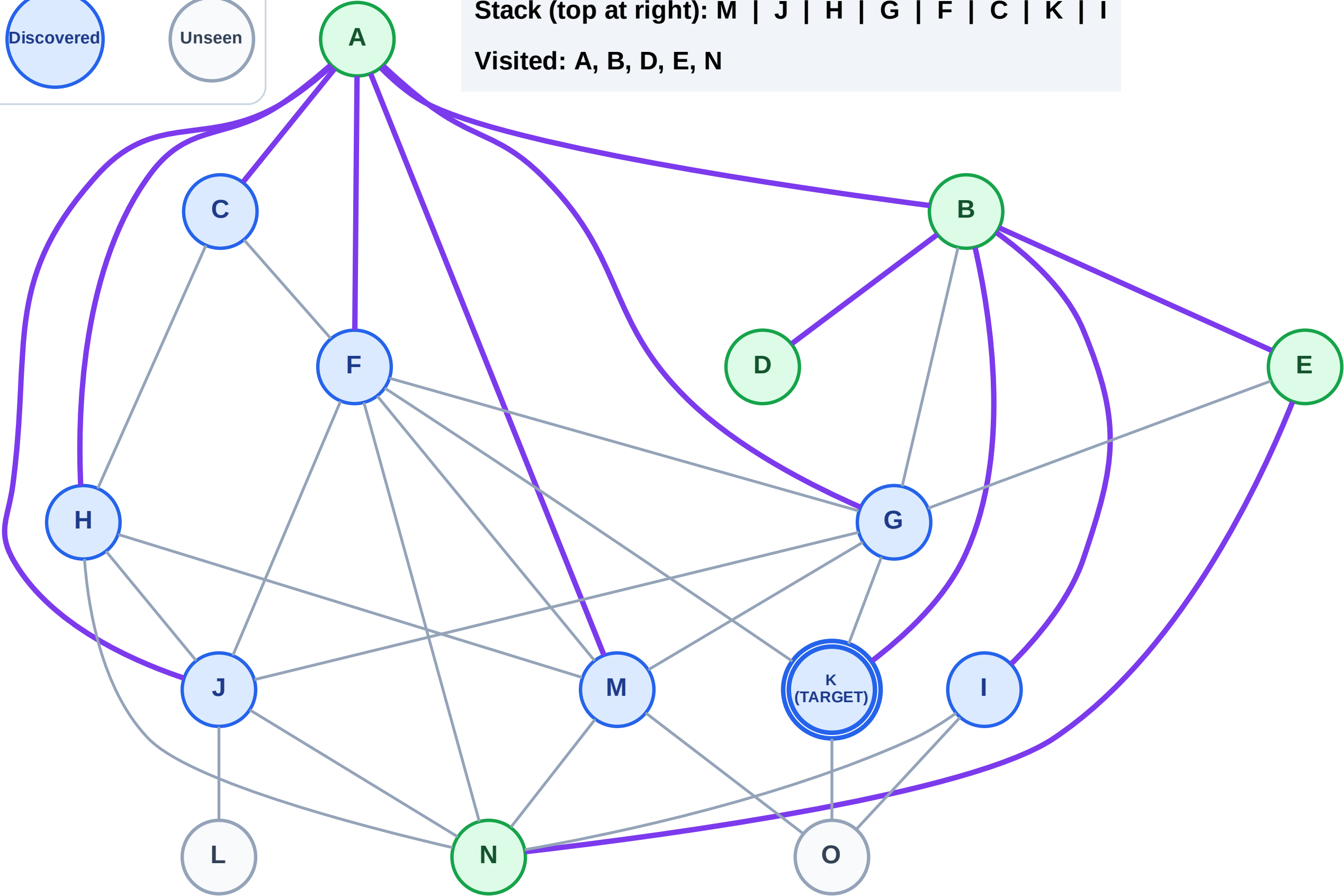
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K | I

Visited: A, B, D, E, N



DFS Traversal

Step 12: Process node I

Legend

Complete

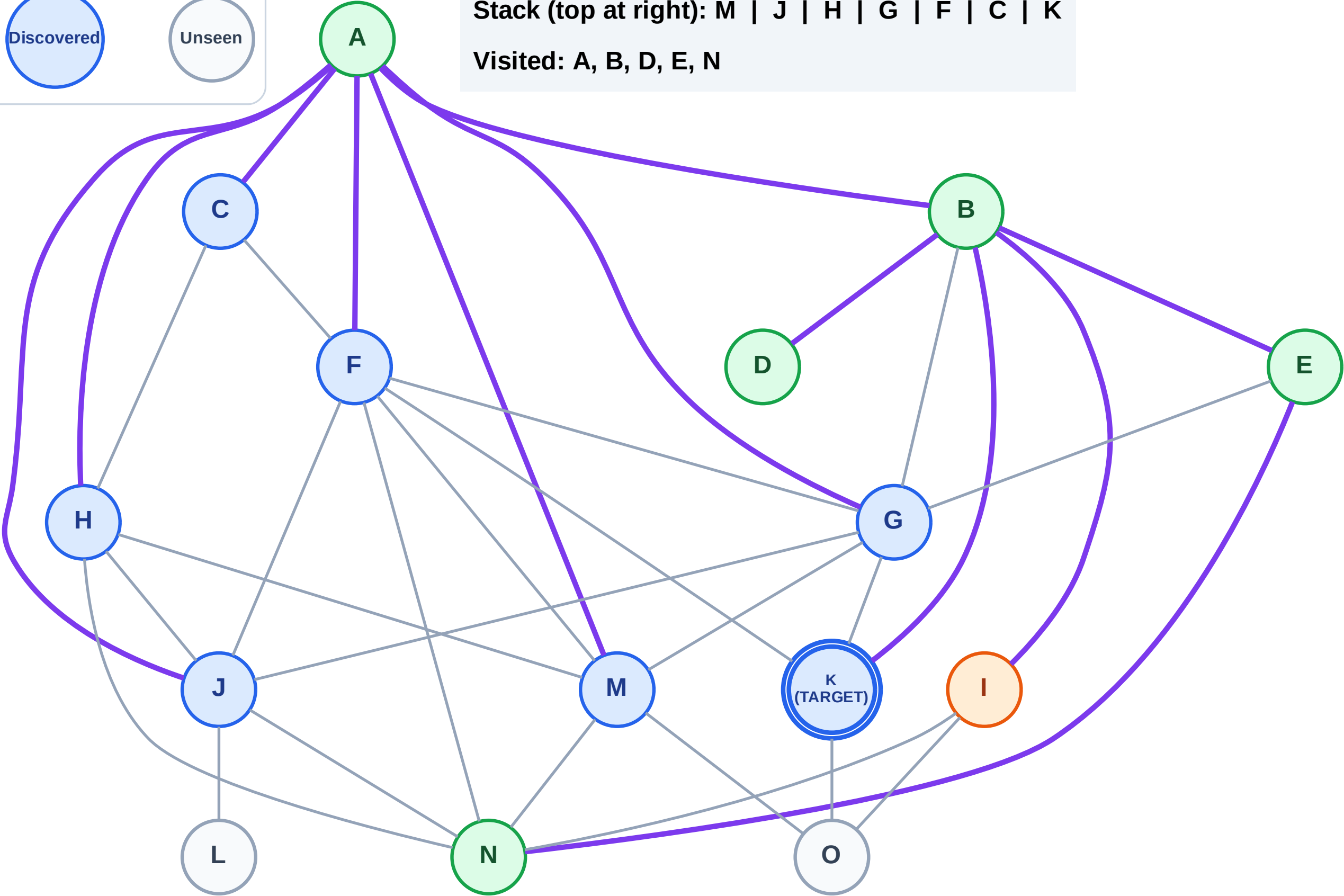
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K

Visited: A, B, D, E, N



DFS Traversal

Step 13: Discover O from I

Legend

Complete

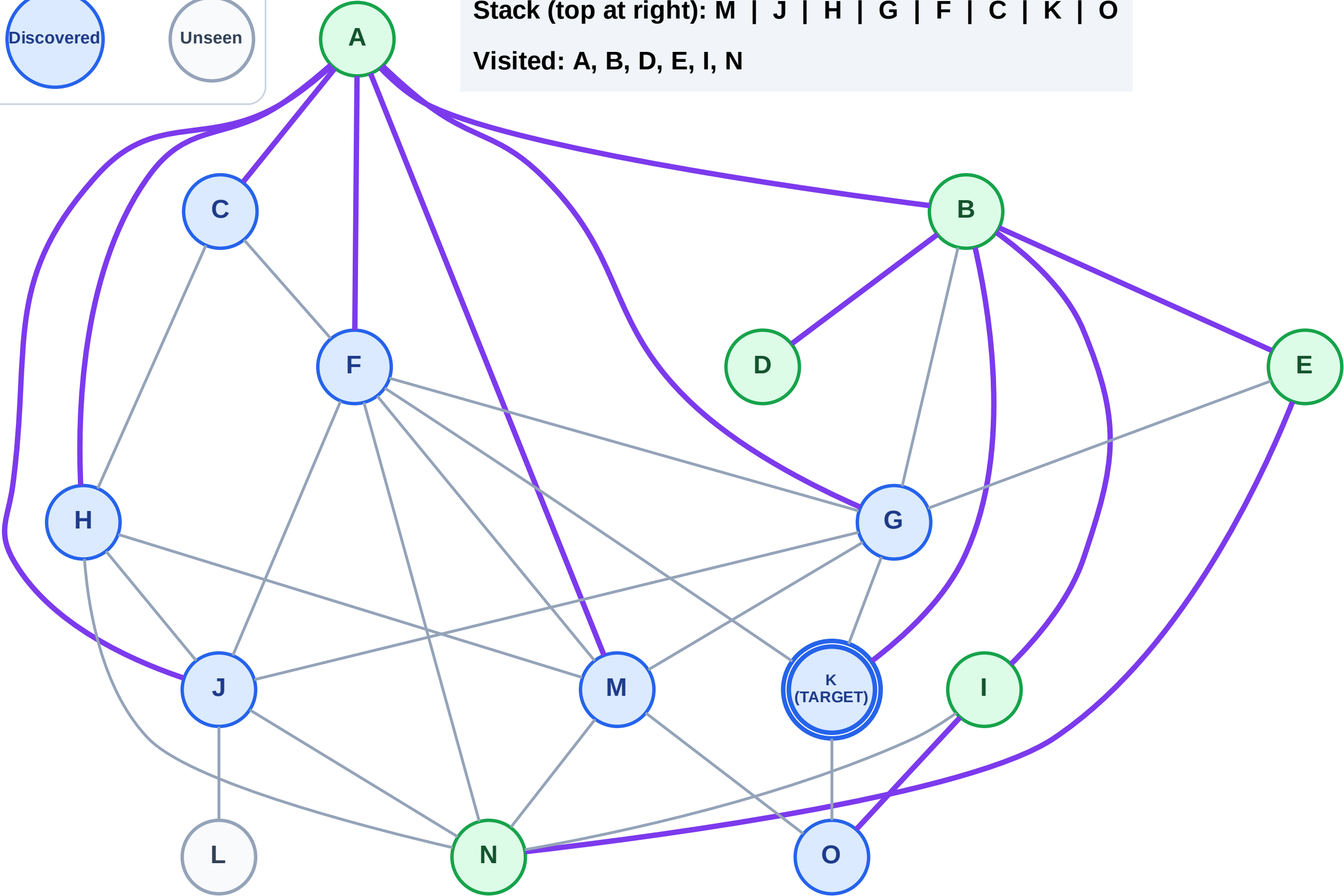
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K | O

Visited: A, B, D, E, I, N



DFS Traversal

Step 14: Process node O

Legend

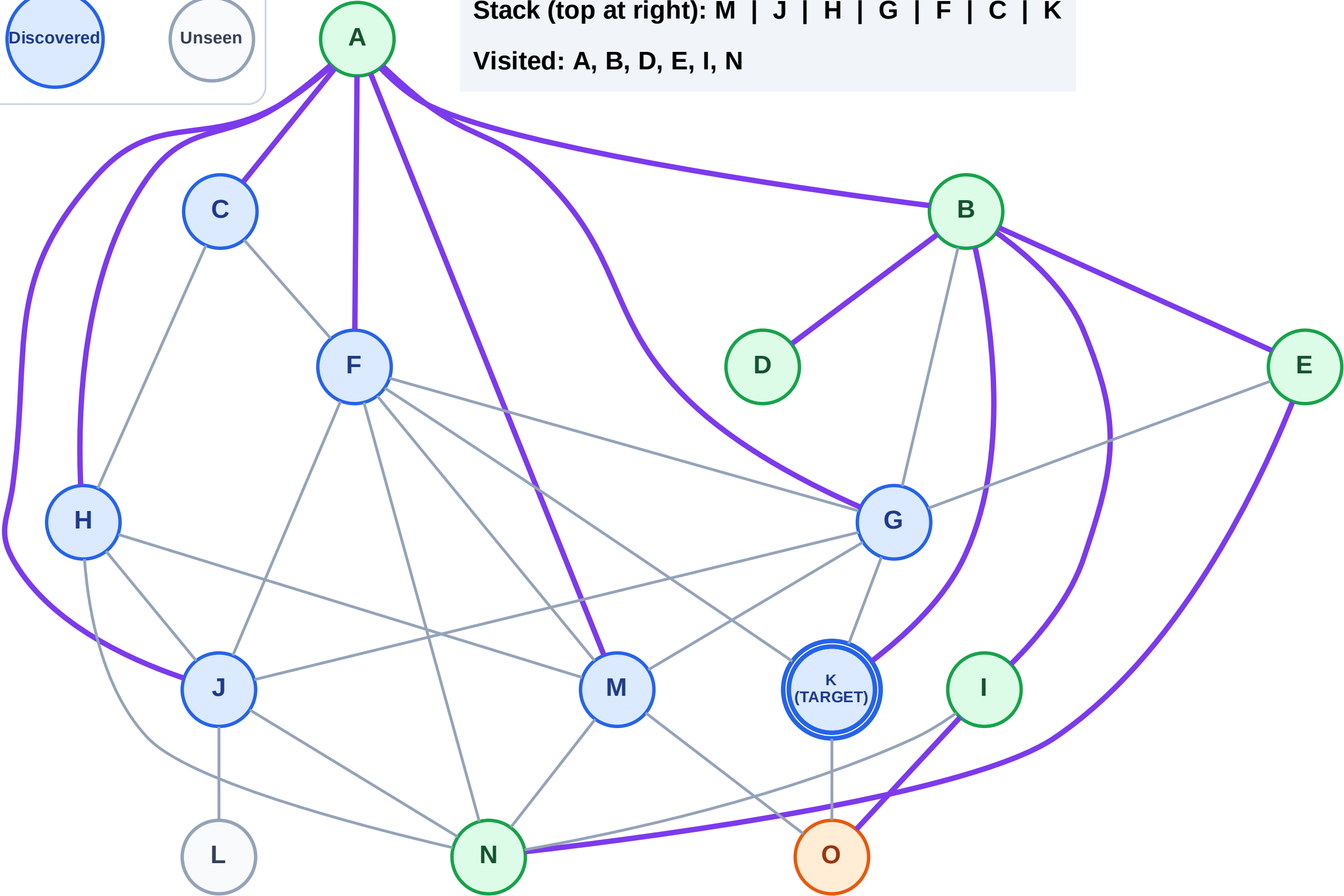
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K
Visited: A, B, D, E, I, N



DFS Traversal

Step 15: No new neighbors from O

Legend

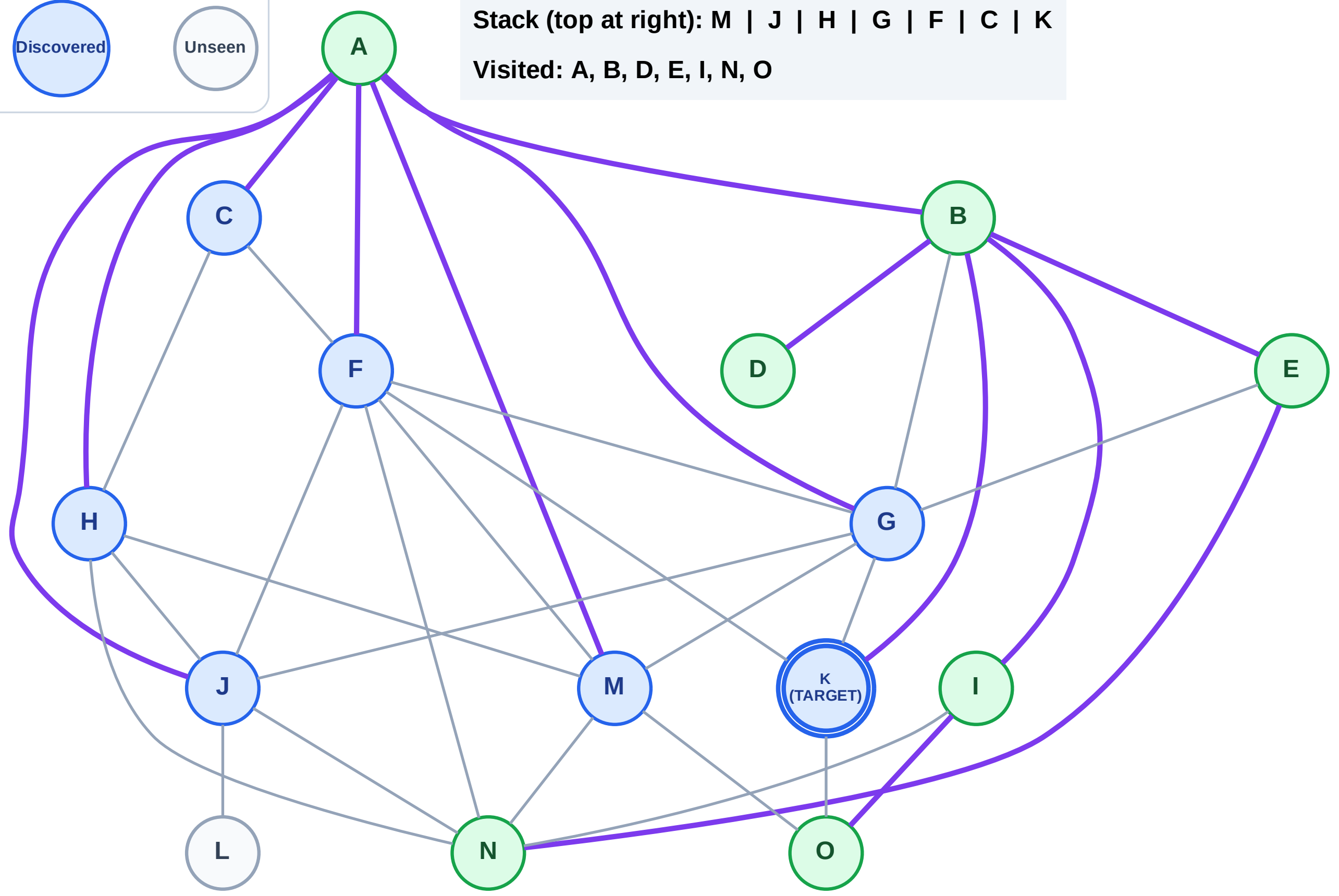
Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C | K
Visited: A, B, D, E, I, N, O



DFS Traversal

Step 16: Process node K

Legend

Complete

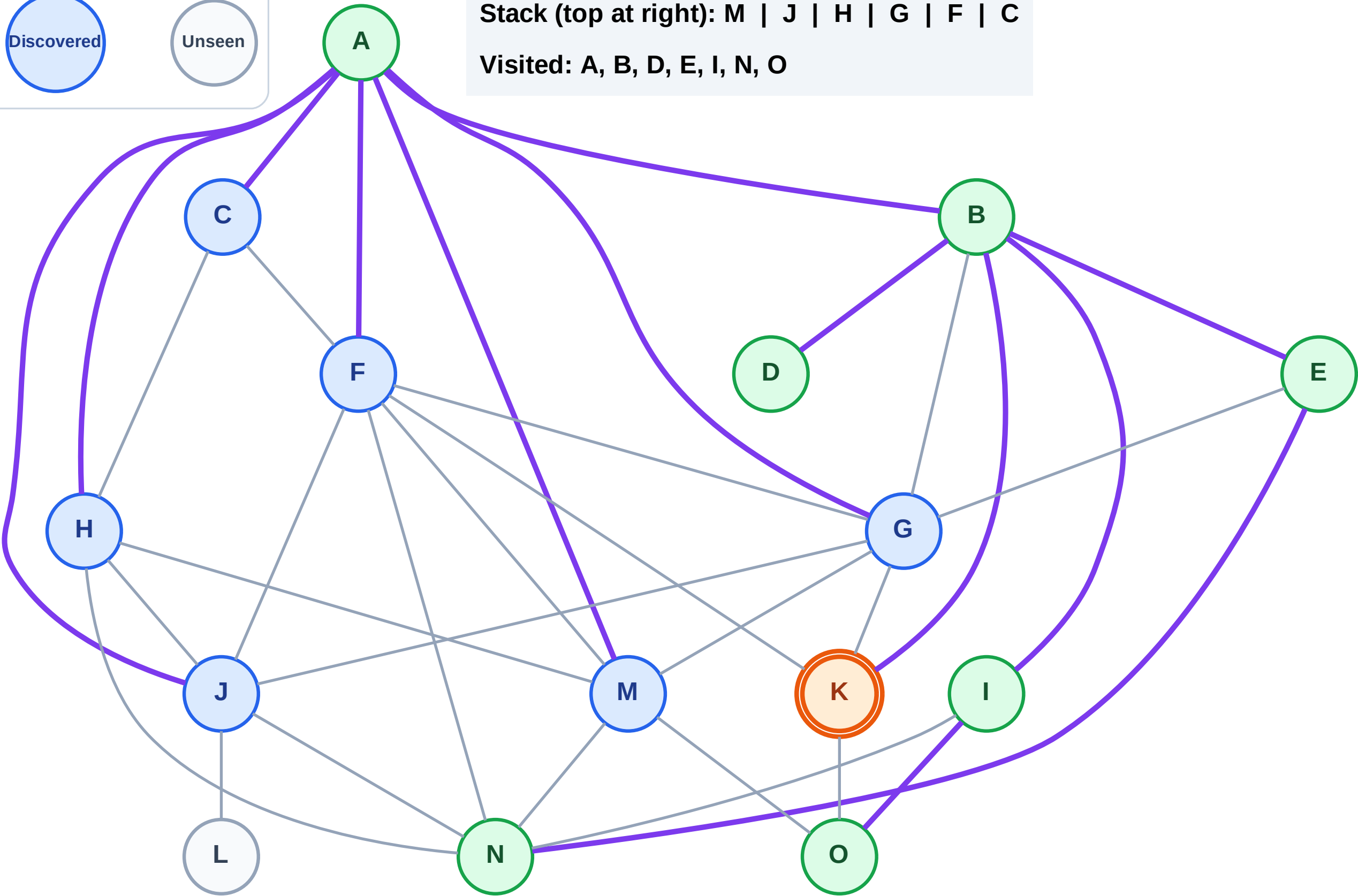
Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C

Visited: A, B, D, E, I, N, O



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | H | G | F | C
Visited: A, B, D, E, I, K, N, O

